Ep #4 | WTF is ChatGPT: Heaven or Hell? | w/ Nikhil, Varun Mayya, Tanmay, Umang & Aprameya

[Speaker 1] (0:00 - 0:09)

To get 50 friends to agree on one date to have dinner is a pain. But that depends. No, if everyone has an AIS system you can reach out to, they can coordinate pretty well.

That's fine.

[Speaker 2] (0:13 - 0:15)

Talk to us like you're talking to stupid people.

[Speaker 1] (0:16 - 0:23)

There's this thing called RISC-GPT. Put on glasses, you're on a date, it just hits GPT and it's like, okay, say these things. And you see it on your AR glasses, and it's like, ah, so...

[Speaker 2] (0:23 - 0:25)

I have Tanmay for that. I ask him what to say.

[Speaker 1] (0:28 - 0:34)

Like a robot is okay with tearing off its leg and hitting you with it. I have the chip in my brain. I'll figure out something else.

[Speaker 2] (0:40 - 0:47)

Okay. Hi, everyone. Welcome back.

So what has Who been doing in the last one month?

[Speaker 1] (0:47 - 0:50)

Well, you've been starring in the second half of Aashiqui 2.

[Speaker 2] (0:50 - 0:51)

Yeah.

[Speaker 1] (0:51 - 0:52)

I love the new look.

[Speaker 2] (0:53 - 1:31)

Well, I was traveling and it was sunny and I thought, half my face is covered. I might not need sunscreen as much. Nice.

That's a good hack. Kind of smart. So if you're in a really sunny place, I was in Phuket.

So if you have a big beard, you can kind of like avoid the sun to a large extent if you wear sunglasses. It's a great hack. It's a great hack, yeah.

Yeah, but I had a good time. I went to Phuket. I spent a little bit of time going to the beach, eating a lot of street food, all kinds of Thai junk food.

Yeah. What have you guys been up to?

[Speaker 3] (1:31 - 1:38)

Well, just been here for a while, traveling. I went to South Africa. That was fascinating.

Beautiful country.

[Speaker 2] (1:38 - 1:42)

And he told me earlier that it's the best place in the world to visit right now.

[Speaker 1] (1:42 - 1:49)

It is. Umang says that after every trip he takes. I just want to see Umang come back once and be like, guys, this country, it's garbage.

[Speaker 2] (1:49 - 2:00)

Please don't go there. We should all welcome Varun. Thank you for joining us on this podcast.

Would you like to like say something about yourself in a minute or something?

[Speaker 1] (2:00 - 2:31)

Sure. So I run a company called Scenes. I also run a YouTube channel called A.V. And I do a bunch of stuff. I've been writing code for 17 years. I've been brought here, I assume, to talk about AI, but I just want to warn everyone, including the people watching, that I quickly switch on Dooma mode. So I'm really pessimistic about what's going to happen to the world in the next 10 years.

So yeah, like if it contradicts with your opinions, feel free to like, you know, just be like, huh.

[Speaker 2] (2:32 - 2:41)

What works for this group is we all believe in totally different things. Nice. Yeah, so you can come with that opinion and I'm sure some of us will share it as well.

[Speaker 1] (2:41 - 3:19)

By the way, I second it. Yeah. No, I'll display the strength because the audience gets scared and they'll be like, dude.

No, no, let's go for it, man. Yeah, rollercoaster. You are on the exponential curve.

And I think there are enough people who are like, this will be an S curve. You didn't say what you were doing in the last month. Doing a bunch of IPL stuff.

I met Ravi Shastri and we shot with him. And it was like being at a, like listening to Ravi Shastri in person is like being at a music concert of your favorite musician because you've heard the voice. So it really felt like that.

He's a, have you guys met Ravi Shastri? No. You haven't met?

[Speaker 2] (3:20 - 3:22)

No, but I've heard that I was asking you earlier.

[Speaker 1] (3:23 - 3:25)

He's a baller. Yeah, don't ask that again.

[Speaker 2] (3:29 - 3:33)

But if he hasn't changed and he retains that youthfulness.

[Speaker 1] (3:34 - 3:55)

We sign off the episode saying Mr. Shastri never change. He was awesome. I think this is my first year seeing the IPL up close.

I've seen the odd match here or there, but I didn't realize when you're this up close, how much of an insane following the IPL has. It's pretty nuts.

[Speaker 2] (3:56 - 3:57)

Do you think cricket is dying as a sport?

[Speaker 1] (3:58 - 3:58)

No.

[Speaker 2] (3:58 - 3:59)

No way, where?

[Speaker 1] (3:59 - 4:00)

Not at all.

[Speaker 2] (4:00 - 4:17)

Not at all. I think amongst the affluent youth. Maybe.

Maybe. Yeah. Because people I talk to like your kids or your daughter in a few years, I think people are switching to football and other sports, especially in that category.

[Speaker 1] (4:18 - 4:47)

No, but it's still an event to go to. Like those kids would still show up at an IPL game. I got out of one game and while going out, you got to punch your ticket in.

You've exited. Once you've exited, you can't come back. There are like hundreds of people outside Chinnaswamy Stadium.

Waiting. Just anyone who exits, asking them saying, can you give me a ticket? So me and my friends, we walked out and this guy, he just cornered me and he said, please give me a ticket.

I said, you can't go back. You can't go. He said, no, I still want to try.

[Speaker 2] (4:48 - 4:49)

How expensive are IPL tickets now?

[Speaker 1] (4:50 - 4:52)

I was in the hospitality section. That was 25,000 bucks a ticket.

[Speaker 2] (4:53 - 4:53)

Wow.

[Speaker 1] (4:53 - 4:54)

Yeah.

[Speaker 2] (4:54 - 4:55)

And how many people in that section?

[Speaker 1] (4:56 - 5:32)

Couple thousand. It's a fairly big stand. A few hundred at least, maybe a thousand.

A few hundred. A few hundred, yeah. The hospitality section is like, it's pretty good.

Yeah, it's like you're in a banquet hall at a five star with cricket just happening to be there and it's an open bar and you have a good time. It's fun. Like, I always used to think that, oh, if you watch cricket in a stadium, there's a sour grapes there, which is like, oh, you won't see anything.

It's so far away. It's such a big ground. But then right there, bro, like it's, you can hear a bat hit a ball.

[Speaker 2] (5:32 - 5:34)

And you have those TV screens there anyway.

[Speaker 3] (5:34 - 6:01)

Right there anyways. But you know, the experience of watching a game and both Chinnaswamy and Wankhede. Wankhede, which is my favorite stadium in India, is because they're tiny.

They're small stadiums. And Wankhede is kind of built vertically around it, right? There's not too much space to sit.

So you feel like you're a part of the game. And same with Chinnaswamy, right? I took both my dad and my father-in-law, thanks to you.

And it was fabulous. Like two 75-year-olds.

[Speaker 2] (6:01 - 6:02)

Were they both Bangalore fans?

[Speaker 3] (6:03 - 6:12)

They were CSK fans, actually. But they just had a great time because, you know… It's a party.

It's a party. The atmosphere was great. They were…

[Speaker 2] (6:12 - 6:14)

Bangalore is always losing.

[Speaker 3] (6:16 - 6:17)

Well, they won today.

[Speaker 2] (6:18 - 6:20)

No, but they've not won a tournament in a long time.

[Speaker 1] (6:20 - 6:21)

They haven't won it.

[Speaker 2] (6:21 - 6:35)

They've never won the IPL. They've never won the IPL. What do you think is going wrong, Tanmay, after spending time with Ravi Shastri and the cricketing crowd?

My cricket analysis. One thing that is going wrong. You can even pinpoint someone's marriage and stuff like that.

[Speaker 5] (6:37 - 6:39)

You're not getting me cancelled.

[Speaker 1] (6:40 - 6:48)

See, as far as I'm concerned, the only matches that most people care about is Mumbai Indians vs CSK-RCB. So, I think RCB is winning.

[Speaker 2] (6:50 - 6:52)

Because they're doing better… They're winning the brand war.

[Speaker 1] (6:52 - 6:53)

They're winning the brand war.

[Speaker 2] (6:54 - 6:55)

They're the largest brand.

[Speaker 1] (6:55 - 7:10)

They're the largest brand by far. So, in case anyone from RCB is watching, you know, and if Dharmesh Chhetri is busy, you can always ask me. It's also correlated, right?

Because they're not winning, the fans are getting… This year, we'll get a cup of Namde. That's the line.

This year, we'll get a cup of Namde every time.

[Speaker 2] (7:11 - 7:16)

And a chip. Usually, you'll get a chip. I would assume because they're not winning, the number of fans will reduce.

[Speaker 1] (7:17 - 7:26)

No, they won't. It's a star-studded team all the time. And there's Virat Kohli.

So, they expect them to win. It is a matter of time. When our time comes, it's the thing that…

[Speaker 3] (7:26 - 7:38)

It was so different, you know, 17th, four days ago, when I was at the game. 70% or 65% of the stadium was CSK fans in yellow. It was like, did I reach the wrong city?

[Speaker 1] (7:39 - 7:42)

That's Dhoni. It was Dhoni. It was Dhoni.

I mean, that was just crazy. It was outside Dhoni.

[Speaker 3] (7:43 - 7:44)

It was outside Dhoni.

[Speaker 1] (7:44 - 8:23)

But the stadium experience is insane. Because in case the four didn't get you enough dopamine, right after a batsman hits a four, Uvantava starts playing. As soon as Uvantava chorus is over, one emcee gets on the mic and emcee starts yelling.

And in the ground, it was pretty funny where the emcee would get on and he would be like, What are you saying, Bangalore? Silence. And then the emcee realizes, oh, I know what to get, what to say.

And he's like, R-C-B. And we were like, three letters, I'll go for it. R-C-B.

It's pretty insane. To your earlier question of, will cricket die? It won't.

Because there are very few moments of celebration and Indians love celebrating.

[Speaker 2] (8:23 - 9:28)

So we're talking about ChatGPT today. To preface our conversation, like, I'm no expert. I'm here to learn from what each one of us thinks about it.

And I think most or many people are like me. We're sitting on the outside. Like, I've watched so many podcasts about ChatGPT.

All the all-in ones, Lex Friedman ones. Even the ones from Elon Musk and the Microsoft CEO. And the Google guys have come back and they're trying to compete.

There are so many different versions of what is happening right now. So maybe we start with understanding what is ChatGPT as a technology. How did we get here?

What happened before? What failed? Was there like a Eureka moment where everything changed?

If so, what was that? So maybe start with defining what is ChatGPT. Would each one of you like to take a shot at it?

[Speaker 1] (9:29 - 10:14)

Well, I think the build-up of data on the internet that's available from all of humanity and what we think has never been as humongous. There's a tipping point beyond which you can actually start building your own human or intelligence. So I think that is what was happening throughout 90s, 2000s to 2010s.

And that tipping point has been reached now where Google results were results, but you still had to add one more level of intelligence to know which link you had to click on and learn from and all of those things. So I think we were all waiting for more intelligent interactions.

[Speaker 3] (10:16 - 10:17)

And I think that's the moment.

[Speaker 2] (10:17 - 10:21)

Do you find the interactions more intelligent now with ChatGPT? Are you using it at KU?

[Speaker 1] (10:22 - 10:39)

Yeah, so we've integrated ChatGPT into KU to help creators. So creators can actually, like you, like there's a button on the creator's screen. The API which connects the ChatGPT.

Yeah, it's an API. So Microsoft, we're working with Microsoft very deeply. So we're making sure that creators get lazier.

[Speaker 2] (10:39 - 10:46)

That's what they want to do. So if I'm creating content on KU, do I have a chat box where I can ask for suggestions? Correct.

[Speaker 1] (10:46 - 10:59)

So on the create screen itself, we've got an idea, like a bulb. You click on that, you can ask for any kind of assistance. You can write a poem to, you know, get me a picture of Virat Kohli or Rohit Sharma.

[Speaker 2] (10:59 - 11:00)

Basically, ChatGPT sits there.

[Speaker 1] (11:01 - 11:11)

Yeah, ChatGPT sits there. And we'll keep building these tools for creators as we go forward. For audio, for video, for pictures.

You know, you can keep getting deeper with it.

[Speaker 2] (11:12 - 11:16)

So if you had to summarize, what is ChatGPT in one sentence?

[Speaker 3] (11:16 - 11:29)

I think it's an assistant to every human. To be a superhuman. As in, how do we access it is going to keep getting better.

[Speaker 1] (11:29 - 11:40)

Right now, it's still an outside interface. It will become a part of us at some point in time. And we will choose to do it or not to.

But I think it's a tool to become super, superhuman.

[Speaker 2] (11:41 - 12:18)

The way I look at it, I mean, correct me wherever I'm wrong, and I will be in many places. But it feels to me like we've been using computers for a long time. Right?

What we spoke, a computer could not understand. And we had middlemen or translation, which we call coders and programmers and stuff like that. The Eureka moment for me in some ways seems to be that the need for the translation seems to have gone away very quickly.

Is that somewhat accurate?

[Speaker 1] (12:19 - 14:58)

I'll tell you the best. I mean, I'll tell you the way programmers think about ChatGPT or the people who made ChatGPT think about ChatGPT. And this is probably not how the layman would think about it.

But ChatGPT is a completion agent. Right? It's a next word predictor.

So if you give it three words, it will pick up the most likely next word. And the best way to prove this to you is and that's not a bad thing. I'm not trivializing ChatGPT's skill set, but I'm just telling you what it is.

Like, for example, I'm going to tell you a statement, fill in the last blank. Nikhil Kamath is a dash. You got a word, right?

Thank you. There's a cluster of words that you would have said or that you would have thought of. But there are words which you wouldn't have thought of.

For example, Nikhil Kamath is a shampoo. That doesn't make sense. The word shampoo doesn't fit.

So essentially what's happening is it's got a probability cluster of what's the most likely word to come out next after you give it three or four or five words. And that's essentially how ChatGPT works. But that's how GPT works.

ChatGPT, and if you've gone to the actual API... What is the difference between GPT and ChatGPT? So GPT is the transformer itself, right?

It's the pre-trained transformer. What is a transformer? So there was a paper many, many years ago called attention is all you need.

So before that, we used to train something called RNNs, which are like very slow. What are RNNs? So RNNs are a type of neural network where essentially they did things in a very different way.

They weren't making... Let's say the sentence is, I am a dog or I have a dog, right? What is the most important word in that sentence?

It's dog, right? Dog is probably the most important. And that's how RNNs work, right?

There's one word that's very important. So when you do things like translation, you want to translate that sentence from English to Spanish, you replace each word one by one. But you know, that's not how language works.

Like if you translate, let's say a statement like mere paas kutta hai to English, not every word will be replaced. Some words like transfer over and stuff. So we realized...

What is the paper called? Attention is all you need. Attention is all you need.

By the way, this is a good exercise. If folks watching at home, sorry to cut you off. You should go on ChatGPT.

I did this two days ago, right? Attention is all you need paper. Can you explain this to me like a five-year-old?

And actually literally does that really well. So they figured out a new technique called attention where they said that instead of looking at a word one at a time, let's look at these as clusters of words. Let's create this...

Think of it like a heat map, right? Of all the words that have a high probability of appearing together and all the words that have a low probability of appearing together. So you basically have this space of all the words, you know, like a word cloud.

But some are together, some are away.

[Speaker 2] (14:58 - 15:09)

So are you in a way saying that the main point of ChatGPT is to be able to predict using probability how to complete the sentence.

[Speaker 1] (15:10 - 17:00)

That is GPT, not ChatGPT. Now if you go to the OpenAI playground And what does GPT stand for? Generative Pre-trained Transformer.

So it's a transformer but somebody's trained it. For a dummy like me, what is a transformer? So a transformer is a type of...

So the best way to think about GPT in general is it's a new type of computer with a new programming language and that programming language is English. Now, if you go to the OpenAI playground, if you look at the first line for ChatGPT let's talk about how you go from GPT to ChatGPT. If you want to create a ChatGPT from GPT, you literally have to...

Your first three lines of your prompt are going to be Hey, you are an AI assistant. You are talking to a human being. Here's an example.

AI colon blah blah blah Welcome, my name is OpenAI ChatGPT whatever. How can I help you today? Human and then human says something.

Then AI, AI says something. Then human dash. Right?

And what you're filling in is that first dash. Before that, there's like three, four lines of prompts

which say, hey, this is an AI talking to a human. So what's actually happening with ChatGPT is it's completing a statement that is simulating a conversation between you and an AI assistant.

Right? In Bing's case, the AI assistant has a name. It's called Sydney.

Right? So what's happening is this completion can be applied anywhere. So, sorry, coming back to what Nikhil said.

What is a transformer again? It's a type of computer. Think of it as a type of computer and you can use transformer in many ways.

But the main way to use transformer was translation. Now we use it for next word prediction. Got it.

And ChatGPT was the transformer. It's a trained transformer. So once you have the transformer, you need to put data into it.

Right? Think of it like a machine and you need to shove as much data into it. So it can predict what would come next based on the data.

[Speaker 5] (17:00 - 17:00)

Right?

[Speaker 1] (17:01 - 17:09)

So someone went and trained this transformer to be a chat assistant. Yes. And that is ChatGPT.

Yes, that is ChatGPT. Right?

[Speaker 2] (17:09 - 17:15)

And you can use this completion in any way. And what is the data that has been dumped into this transformer GPT?

[Speaker 1] (17:18 - 17:30)

It's all of the web. But it's mostly Reddit. Right?

And the best way to understand... Mostly Reddit. Yes, it's Reddit.

Anywhere... The internet was mostly forums. The more UGC, the better.

[Speaker 3] (17:30 - 17:31)

Yes, the more UGC, the better.

[Speaker 1] (17:33 - 18:20)

And I want to mention this because people get this wrong. It's not really like whatever question you're asking has been asked before. It's more like with any AI, it's learning the underlying pattern of how this conversation happens.

The best example of this is when you fine-tune ChatGPT. You can use ChatGPT as a base prompt and response. I'm going to give you many, many pairs of it.

Maybe 10,000, 20,000, whatever. And I can modify ChatGPT. Now, if you do it with your...

Let's say I take your WhatsApp and I just rip through all the messages you've replied to. So the message that's been sent, messages that you've replied to. And I just dump it into ChatGPT.

Eventually, it'll also pick up things like your spelling mistakes. It'll pick up things like your style of speech. Right?

So it's somehow started learning the underlying patterns of what?

[Speaker 2] (18:21 - 18:33)

Of the text body that you've given it. Wait, wait. I'm still a little confused.

So GPT is a transformer. It's a training transformer. You dump data.

It makes predictions based on the data.

[Speaker 1] (18:34 - 19:41)

So transformer plus data equal to GPT. Think of it like that. I'm obviously simplifying it, but that's how it works.

And when you say training, what does that mean? You feed this into the transformer and you say, well, let's say if I have 100 pieces of data, I'm going to keep 20 pieces of data aside. I'm going to be like, these are the prompts.

These are the response. I'm not going to feed this into the transformer itself. All this other content, I'm going to feed into the transformer.

Now it's going to learn the underlying patterns. It uses a neural network. Give me an example.

So let's say I have a statement called I'm a dog. Let's take a better example. There's some data where somebody has responded to that statement saying something.

Over time, if you feed enough examples, it starts understanding that Varun is the least important keyword here. And it'll start understanding that this person just wants a diet plan. Or this person

just wants to know what to eat.

So here is the statement. It's generalizing in a way. So once it's learned the underlying patterns, we take the other 20 pieces that we had kept aside and we tested on this.

[Speaker 2] (19:41 - 19:42)

Another digression.

[Speaker 1] (19:42 - 19:52)

How is it learning? That's complicated. It's a neural network that's behind the scenes.

Even experts in the world don't know exactly how it's coming out with enough response.

[Speaker 2] (19:52 - 20:09)

I know a little bit about neural networks because we use it in trading. We dump in volume, price, time, data, and we wait for the network to throw out patterns that it could recognize in the past. But when you say learning, that part I'm a little conflicted about.

Yeah.

[Speaker 1] (20:09 - 21:07)

So the best way to think about it is every piece of data or every piece of content is a piece of data. So the best example of how I can explain to you outside of text is maybe let's say faces. Let's say I want to train your face.

What is training my face? So maybe I want to copy your face. Maybe I want to put your face on Tanmay's face.

Now what's important in a face? By training you mean learning my face? Yes.

Learning the features of your face. So we go into features. Now how do you define what features are?

You might say you have a nose. Nose is important. You have eyes that are important.

But then there are also other things that humans don't understand. We don't understand these patterns. For example, the distance between the corner of one eye and the nose may be important.

We don't know if it's important or not. And the reason we deploy these ML models is because we don't know the pattern well enough. We don't know the underlying reason why consciousness comes out or why somebody says something or how people form sentences.

We still don't know all of this.

[Speaker 2] (21:11 - 21:31)

So let's summarize. Let's go part by part so we remember what you've said so far. You're saying so much.

So GPT transformer. Data is dumped into the transformer. It learns from the data which has been dumped to be able to predict the next word based on probability.

[Speaker 1] (21:31 - 21:32)

Yes, based on probability.

[Speaker 2] (21:32 - 21:38)

So these guys took GPT added the chat bot function to it created chat GPT.

[Speaker 1] (21:39 - 21:59)

There's a transformer. Then they trained it to be an AI assistant when someone chats to it. Yeah, basically what it's doing is it's in completion mode and it has many other modes which are not as good.

For example, it has an insert mode. It has a edit mode. But we don't use all of that because it turns out completion is great.

But till now we've been trivializing it.

[Speaker 2] (21:59 - 22:05)

We're very stupid people. Let me preface. Talk to us like you're talking to stupid people.

[Speaker 1] (22:11 - 22:55)

I'm going to restrain myself. Talk to me. So let's say this.

The best way to think about it is GPT or chat GPT is a new type of computer with a new programming language and that programming language is English. Plain English. And you're right.

Fundamentally we were talking to computers with code. With code you are very verbose. You need to be very specific about what you want.

And if you miss a word in the syntax nothing, the computer is not going to understand you. Now we have a way to express in English or any other language because it turns out the underlying

patterns of all languages are similar. And there are many papers about this Can I ask a question again?

[Speaker 2] (22:55 - 23:06)

Why was it so hard up until now? I'm talking about why was it so hard to make a computer understand instructions in spoken language?

[Speaker 1] (23:11 - 23:55)

Because if you take something else away from it he might take something else away from it. So it's open to interpretation. And with a computer the reason you're so specific in code is because a computer is stupid.

It's smart but still stupid in some ways. So it can and even now with tools like AutoGPT it can go on a tangent. It can go wherever it wants.

So as specific as you can is good and till now the only way we could be specific was through code. And you have plenty of coding languages and you have multiple let's say the languages we use today especially in web technology there are many levels of abstraction below it. So now we have a way for layman to talk to a computer and make it do what you want because it now understands you.

[Speaker 2] (23:56 - 24:57)

But how far does that go? Say for example I use a program to conduct a certain function which up until now a programmer who is a colleague of mine would do for me. Let me give you an example of a program.

Let's say I run an investment strategy which trades based on correlation between Indian markets and Hong Kong because historically they are very correlated. When that goes up somebody buys here when this goes down somebody sells there something like that. So if I had a programmer build that for me integrate between data vendors in both the markets figure out a way to reduce the latency of the strategy so the orders get executed fast all of that if I need ChatGPT to either help me out or replace the programmer in that process which is very specific to me what do you think are the

[Speaker 1] (24:57 - 25:05)

So mostly in this particular use case you have programmers counting on APIs or scraping. If there are no APIs available you are probably going to scrape a page.

[Speaker 2] (25:11 - 25:29)

In return for that they give you data at a certain periodicity So you call them different levels of

data like level 1 data you will probably get 1 tick a minute if you are buying level 5 data it will give you like 10 snapshots in a second So the exchange does it and you speak to the exchange through an API.

[Speaker 1] (25:30 - 26:03)

So essentially you could say in this particular case your developer is a plumber he is stitching a bunch of APIs together there will be a point where you can just take because ChatGPT might not have been trained on the data of that exchange So there is a way for you to just take the entire documentation of that data and today after this hopefully I will show you this you can take it, you can just dump it and you can be like this is the documentation please write me code to do whatever you want

[Speaker 2] (26:03 - 26:33)

But can I tell you I tried this So I went into ChatGPT correlation strategy between X and Y and I also asked if it could check what two instruments are more correlated based on past data But the problem with anything finance and economy related the one big hurdle is it doesn't have data after 2021 So that immediately negates

[Speaker 1] (26:34 - 27:41)

That's a temporary hitch because the fact that you can't put in data it's because ChatGPT for GPT 3.5 is like 4096 tokens and then for now a 32k token window You have to explain each of these things How many words can you write in the window You can't just put in a 50 page document, it will cut you off after certain pages and be like it's too long There is a version of GPT 4 where now you can dump in up to 32,000 tokens Tokens are words? It's correlated with words So there is a ratio between word to token which is somewhat different Now what ends up happening is if I have this content of the API docs of exchange 1 API docs of exchange 2 I need to tell ChatGPT this Or better yet, allow ChatGPT to Google so that I don't even have to tell it this So there is a new layer on ChatGPT called ChatGPT plugins which allows it to hit things like search It will search by itself It will think and then it will be like I need to use this endpoint like this

[Speaker 2] (27:42 - 27:47)

Is it ok if I move my chair this way? Because my neck is paining otherwise You are sure right?

[Speaker 5] (27:49 - 27:49)

Yeah?

[Speaker 1] (27:53 - 28:54)

Ok So essentially what's happening is You are right You don't have to cut that part You can let it

play There is a lack of data in terms of what specifically you want to do I would say most technology at the edge will require some specific piece of knowledge If somebody can write that piece of knowledge then you can do the rest Not ChatGPT itself You probably need to use a tool like AutoGPT Because you need recursion You need the ability for ChatGPT to say See every decision we make From the CEO and below If you look in the org chart It's like you make a decision You delegate some of those decisions So let's say somebody working under you That person delegates 4 decisions to people under them And all this happens in parallel And the output of person C Who works under person B And then back to you So AutoGPT Tools like AutoGPT Allow ChatGPT to kind of have memory Allow you to use external documentation Allow it to hit tools like Google

[Speaker 2] (28:54 - 28:57)

What do you mean when you say Allow them to have memory

[Speaker 1] (28:57 - 29:03)

So right now ChatGPT doesn't have memory If you close the page And you come in, it's gone

[Speaker 2] (29:03 - 29:27)

No, it remembers So if I have spoken to ChatGPT About 10 different topics On the left If I go back to that particular chat It's still there Let's say I have a chat About a certain topic When I go back there And ask it a question It already has context From my previous conversation

[Speaker 1] (29:27 - 29:49)

So it has memory but in a short period of time Until you are in that session Or if you go back to that session Even after days it has it But what I am saying is Imagine you asked You remember it has these multiple tabs on the left So what I am saying is What if it remembered through all those conversations Because you need many parallel instances of ChatGPT

[Speaker 2] (29:49 - 29:52)

You are saying if it's able to weave In between all those conversations

[Speaker 1] (29:52 - 29:58)

And realize all that you are thinking We have long term memory AutoGPT does it in a slightly different way It uses something called vector embeddings

[Speaker 2] (29:58 - 30:06)

Can you differentiate Can you again give us two dumb lines on AutoGPT We figured out what is ChatGPT

[Speaker 1] (30:06 - 30:20)

It's got memory It's ChatGPT with memory It's ChatGPT with the ability It has a slave And it says you go do this And give me the results So there is a master and there is a slave And you can have multiple masters

[Speaker 2] (30:20 - 30:38)

It's like an org chart So AutoGPT is differentiated By having long term memory And being able to weave Through the different topics That somebody might have spoken about And delegation And also does it work In a manner where It picks off the data

[Speaker 1] (30:44 - 30:47)

It's not about intelligence per se But it's just about communication

[Speaker 2] (30:47 - 30:55)

But then why do we need A separate AutoGPT If we have ChatGPT Wouldn't this evolve into that

[Speaker 1] (30:55 - 31:03)

Absolutely So AutoGPT was made by some random guy On Github It doesn't even have a real name It's some Gravitas or something

[Speaker 2] (31:03 - 31:08)

So if ChatGPT adds long term memory And is able to scrape data Of other ChatGPTs

[Speaker 1] (31:08 - 31:22)

It is AutoGPT So AutoGPT is a proof of concept More than anything Because it still gets a lot of things wrong But what I find the coolest part About AutoGPT is It can use any Python script in existence And it will find out What script it needs and download it

[Speaker 2] (31:22 - 31:24)

It gets that from Github

[Speaker 1] (31:24 - 31:43)

There is a central Python repository And it will just be like I need a tool to clip videos It will go through the repository And find me a repository That sounds like it can clip videos It can go to all Python repositories So I once used AutoGPT But tell me this

[Speaker 2] (31:43 - 32:07)

Whenever I get that ChatGPT In itself is an organization It's not a not for profit It's a company, it's a corporation All these places That it is going to learn Going to pick up data This Python repository Twitter, Reddit like you mentioned Why are people allowing For another corporation To benefit from them

[Speaker 1] (32:09 - 33:05)

We saw this in art There is a website called ArtStation And ArtStation for the longest time They had all the 3D artists come there They had all the Unreal Engine guys come there Put up really cool 3D artwork 3D and 2D artwork And for a long time they laughed at AI AI is never going to be able to do what we do Then Mid Journey started training on them And many other image models Image models use something called diffusion And they started training on them Then what happened is These guys said how can you use our data And Mid Journey's counterpoint was this We are not copying your data We are learning from it I am not taking your data As is transmuting it here and there And putting it out I am learning the underlying patterns Between you made a face I am learning the features of this face And I am not hand coding all these features The AI is learning it by itself Just the way our human brain learns it You could close your eyes And you could figure out You know in your head

[Speaker 2] (33:09 - 33:14)

Can I draw an analogy to something That is more tangible in my mind Say songs

[Speaker 1] (33:14 - 33:16)

You can get inspired by songs

[Speaker 2] (33:16 - 33:32)

So say there are songs Which are owned by different labels There is song X Are you saying If I were to go Pick up the song Copy a certain part of it Use it in my song

[Speaker 1] (33:32 - 34:17)

No but you are not copying a certain part of it You are learning the underlying pattern of the song If I say it is a country song I am learning country songs Eventually I can come up with something brand new That sounds country But is not really any of these songs And it is exactly the way The human brain does it Well not exactly but in a similar way If I show you a face And give you a piece of paper And say draw a completely random face You can actually do it You would be able to do it And none of us would be able to recognize What that face is So that is what ChatGPT is doing They are not reproducing your data in any way They are just learning from it And the reproduction rate for something like Stable diffusion is 1% What does that mean?

[Speaker 2] (34:17 - 34:20)

What is the reproduction rate?

[Speaker 1] (34:20 - 34:37)

Let's say I have trained it on 10 pictures of your face You have learnt 10 different pictures of my face 10 different pictures of your face And I say give me a face of Nikhil Kamath Most of the time 99% of the time It will show me a new picture of Nikhil That doesn't look like any of the other 10 pictures But has learnt the features of distance between

[Speaker 2] (34:39 - 34:43)

But isn't that at some level Learning and copying is the same thing?

[Speaker 1] (34:43 - 35:31)

If you do that then you have to sue Any company that Essentially decides to build Like an exchange like yours You would have to sue anyone Who takes inspiration from you In art that happens all the time Like all anime or all manga take inspiration From each other for example You can't go around suing people And saying you took inspiration Like DMX once tried this The rapper DMX Like a rap Like a dog sound he keeps making And some other rapper copied it And he went to court and said that's my style How can you copy that?

And got thrown out So inspiration, fair use You can take a thumbnail You can take somebody else's picture And absolutely modify the hell out of it Such that it doesn't look like it at all Everything is derivative work

[Speaker 2] (35:31 - 35:42)

All of life is plagiarized Everybody is inspired by something There are probably only So you pick any writer Anything in anything

[Speaker 1] (35:42 - 35:50)

No the 10 unique things I would argue Are also not original They are picked up from the environment You just look at whatever the sky The most unique packaging

[Speaker 2] (35:50 - 36:00)

So I think the question really is That temperance in a way What level is okay And what level is not okay And I guess that is on society to define

[Speaker 1] (36:00 - 36:29)

The problem I have is that If a human gets inspired by something That's okay But an AI can crawl through the entire internet And be trained on the entire internet And beat everyone So my worry

is Can you arbitrarily put a rate limit And say well you can only learn at this rate Otherwise you are going to get sued So you can't do that Our legal system is not ready for this And we don't want to launch lawsuits Against anyone who got inspired by another person

[Speaker 2] (36:29 - 37:06)

So let's summarize What you have said till now So there is some kind of a flow For dummies like me You said chat GPT GPT is a transformer It consumes data Helps in making predictions Using probability You have made it a chat bot in a way Called a chat GPT Then there is auto GPT Which has a hierarchy Where it can communicate With different chat GPTs And one level higher Decision making Is that right?

[Speaker 1] (37:06 - 37:44)

Long term memory It has access to tools What are these tools? Anything You want a Python script How do you download a YouTube video? Why can't I do that in chat GPT?

If you tell chat GPT to download a YouTube video It will tell you I can't do that It can pull a Python script But chat GPT can't execute there In fact there is a plugin on chat GPT Which will probably enable that There is one with code execution But otherwise auto GPT will go to Google Find the best YouTube to MP4 downloader Put your link in there Download it to your local drive And extract whatever it needs to

[Speaker 2] (37:44 - 37:57)

Why can't chat GPT execute Say for example Google Python programs It throws it out It says do this, do this Run this, do that But it doesn't execute Why can't it execute?

[Speaker 1] (37:58 - 38:30)

It doesn't have an execution layer It is just outputting text right now And it can output markdown as well So markdown is like HTML Chat GPT can spit out HTML HTML is like a Think of it like a Templating layer for the web You want to display something on the web You want to put something inside tag You have a body, you put that inside the body tag And then you can style it with another thing called CSS So what I am saying is Chat GPT can output all of it It can do Python code as well But it can't execute Python code

[Speaker 2] (38:31 - 38:35)

But what will auto GPT have That chat doesn't

[Speaker 1] (38:36 - 39:29)

It has a terminal What is a terminal So a terminal is like your command prompt It has access to your command prompt It will probably have access to Python And it can execute that code It can

find the Python package Download it by itself And then say I am executing it Why didn't chat GPT enable that from day one Because it would be insane to have I think the reason they haven't Released plugins Because they have this plugin The code execution is a plugin Which is responsible for breaking Or jail breaking chat GPT Now if you give chat GPT access to the internet And give it access to all tools All hell could break loose But auto GPT broke all hell Because they are using OpenAI's API key Basically putting your own API key So using the intelligence from there And then it is using its own set of tools Saying we will go to Google Find the right tool and then execute it

[Speaker 2] (39:29 - 39:34)

But maybe you can come in here And say what do you think of chat GPT What is it to you

[Speaker 3] (39:36 - 41:26)

Anyway define chat GPT Like we all did I mean we started using it Within Daily Hunt To start generating content Around our Espresso feed And it is incredibly useful So I think there are utilitarian jobs That are being done Very very effectively By the product Everything that Varun just said It is generative It is going to help generate What is relevant I don't want to go more Into the technicality And you talk about Embeddings and everything Varun is the expert But I think if you just abstract it One level higher The world is full of stupid people Right If you just abstract it A little way out Right Basically AI was Till 2012 or 14 It was academic Research Suddenly for the first time It has come to life Computers have got So much more powerful Because of this new Let's just call it Language technology Whatever you want to call it There are no rules of engagement And you were kind of going there In a way And what it could do to society If kept in the wrong hands Or if used for the wrong things Varun will start now Highly destructive Like I can't tell you How crazy it could be You could alter You could alter I don't know if we want to go that open In this session Of course, that is the whole point You could start another war You could

[Speaker 2] (41:26 - 41:31)

How do you presume A war can start Using chatGPT

[Speaker 3] (41:31 - 41:39)

Well it was involved In the current Ukraine crisis as well They made that Deepfake of Misinformation

[Speaker 2] (41:39 - 42:06)

But I think that's the indirect way But don't you think That beyond a point Will organically inculcate Some kind of curation Where we will only consume Information from a trusted source And anonymity which chatGPT will bring When it's trying to create Misinformation won't work Maybe in

social media it will Vehicle where one Usually consumes news

[Speaker 1] (42:06 - 43:34)

So I have a theory on this I don't know if it's the right theory Or wrong theory We have an immune system And let's say You are now exposed to covid For the first time You are going to have a violent reaction Your body is going to fight it But your immune system is such That you don't have the same Reaction to something you eat You drink coke You don't have a reaction to it That's because your body Has a sense of what's safe What's not safe And if it finds a signature That's not safe It fights it It creates antibodies against it So that the next time it comes It like violently destroys it immediately I feel the brain has an immune system That's similar And that's why we do things Like we get politically inclined There are a bunch of ideas We just shut it out I feel like Way to tell people A counter argument Or show them the light Whatever narrative you are trying to spin In the pinhole of their current Immune system So I can show you why the right is good Even though you might believe in the left And there is a specific set of words You can do that with And I feel like ChatGPT will eventually Get good enough It will get good enough For your brain When it sees a video The first thing you are going to do Is believe it's real Even though ChatGPT exists

[Speaker 2] (43:34 - 43:36)

Will that change because of all this?

[Speaker 1] (43:36 - 44:18)

Tomorrow if I see a video I will not believe it's real Do you believe that Most of the text on Twitter Right now is real What do you mean by real? On your feed Let's say there are 30 people On your feed And maybe you followed 30 interesting people How do you know that the content is real? You are right in the sense That you trust those people And you know those people are appearing Because you are giving them a follow Now the thing is That's a first party Where I am putting out my own thoughts But most thoughts outside of social media Are a second party Talking about a first party For example in news They can tell you something about you And the audience will believe it

[Speaker 3] (44:19 - 44:21)

I mean it's just going to the point That we were having earlier

[Speaker 1] (44:21 - 44:26)

Because fake news has existed for a long time But people still believe fake news on WhatsApp

[Speaker 2] (44:27 - 45:05)

Some people believe fake news on WhatsApp Let's assume Some is a bigger number than some

And it's like I wouldn't say majority But a reasonable percentage of people But going back to what you were saying About Twitter The reason I follow those 30 people Is because I believe At some level I know them What they are putting out there Is authentic Because I have some kind of a relationship with them Or they are verified accounts of people In a position of You know for some reason They are intelligent Or I appear to think so How will that change?

[Speaker 1] (45:06 - 45:24)

Unless you kind of like Get them to start putting fake stuff out No that's layer one If you follow 30 clean people Or in real life Nothing is going to change But remember they are also vulnerable To watching the news At the end of the day But that's happening already Why does ChadGPT change that?

[Speaker 2] (45:24 - 45:29)

There's fake news in the world already All of us are experiencing it In one way or another

[Speaker 1] (45:30 - 46:04)

There's this thing called the Ash experiment You get 10 people in a room And you show them a graph Or you show them a number Let's say 9 And you ask them what number is this And everyone says 10 And you are the last person to be like This social pressure of conformity Like The actual experiment is you show two lines And you ask them which line is taller And for most of the experiment Let's say you show them 10 slides 9 slides Let's say thing B is taller And everyone says B, B, B And the last one B is shorter but everyone else still says B is taller Then what do you do? You don't look like an idiot So you're like I'll conform

[Speaker 2] (46:04 - 46:29)

What if I take the other opinion Where once For the lack of a better way of putting it Once the quantum of fake news And persuasion towards incorrect stuff Increases significantly Human mind will organically learn To disregard I don't think we'll be able to tell A higher number of it I don't think we'll be able to tell

[Speaker 3] (46:29 - 47:11)

See my view is It has a larger impact So you start with fake news You can manipulate opinion You can manipulate economies Would you agree You will dispense of Tons and tons of jobs If you just look at the job Hierarchy in the market Not everyone is at that level of intelligence Where they have Jobs of a very thinking nature Sure you can argue that It's mostly inefficient jobs Maybe I'm just pained By what I saw in South Africa But when I look at countries Which have 33% unemployment rate Or 40% unemployment rate Even India has I think 7-10% if I'm not mistaken

[Speaker 2] (47:13 - 48:03)

Each country has a different rate But it is The way they calculate it Is different So when they say unemployment They mean above the age of 18 Educated, able, looking for a job Above the age of 65 Maybe the world evolves in a way Where all predictive models Fed similar kind of data Will throw out generic outputs Not necessarily Like if you say movie script And you base it on the data Which is freely available online At some level It will learn of the movie scripts out there So at some level They will all be in a certain category So maybe the future Is for nuance Coming out with ideas That have never been done before Or not never been done before But not

[Speaker 1] (48:03 - 48:04)

Those would be cross domain ideas

[Speaker 2] (48:04 - 48:07)

Which have not been the norm In recorded history

[Speaker 3] (48:08 - 49:37)

I tend to look at this Very differently So I was reading this New York Times article Which came out yesterday And the title of the article is Will this be the Indian century Four key questions Without getting into the details He's saying The only certainty about The biggest country in the world Is that it will be Unlike Any that came before it So he's basically Talking about The century could be India's century He's asking that question And he ends the article with a line He poses four questions What are the questions?

The questions are really around We are the largest democracy Will our demographic dividend Actually pay off Right? And the point that he tries to make Is for this to pay off We've got to be creating 10 million new jobs a year Okay? And he ends the article Saying the only certainty About the biggest country in the world Is that it will be Unlike any that came before it It's going to be different He doesn't have the answer He's asking the questions The thing that worries me Is if you look at Today, what's the biggest problem In the world?

And I know I'm talking to you Being a capitalist Probably the wrong thing to say But capitalism is broken Would you agree?

[Speaker 2] (49:38 - 49:40)

Capitalism might be broken It's totally broken

[Speaker 3] (49:40 - 49:42)

But it's better than the alternative

[Speaker 2] (49:42 - 49:43)

Yeah, I agree

[Speaker 3] (49:43 - 49:45)

Hold on One of the alternatives

[Speaker 2] (49:45 - 50:38)

History has taught us We've had socialism We've had communism Many people talk about this I feel Forms of governance Economic models Are so cyclical We've seen this time and time again If you go way back in history After capitalism Plato says this very well When he describes democracy But after capitalism It's followed by Some nutcase Who's like offspring of his At some point of time Who comes out and is batshit crazy Followed by revolution again Followed by people Wanting power back in their hands again Which is again a form of democracy And then that cycle repeats all over again So the cycle is natural Like everything in the world This is cyclical as well

[Speaker 3] (50:38 - 50:50)

So if you think about the world That we're living in And why do we have bad actors In society It's because The largest percentage of wealth Is concentrated in Very very very small pockets

[Speaker 2] (50:50 - 51:42)

I hear this argument so much But I can make so many Arguments to portray That the world we live in Right now under capitalism Is the best version of the world Not just for the affluent But for all sections of society Like you go back 60-70 years Our average age used to be like 40 I know what you mean We're living in really really good times Now we can complain about The issues of capitalism And I think there are many And I also feel Capitalism has to evolve in a manner Where the anomalies of capitalism Like all of us and many others Will have to become more benevolent Not necessarily So it's compassionate capitalism Not necessarily for society But to continue this system For them and for others

[Speaker 3] (51:42 - 52:10)

So let's Now superimpose Where we were And what Varun was talking about Assume that the power Of what this engine has Is concentrated In the hands of A few And imagine that it is capable Of doing what Varun described One is generating Tons of content That looks real That isn't real Shaping public opinion

[Speaker 1] (52:10 - 52:12)

That degree of power is insane

[Speaker 3] (52:12 - 52:20)

Degree of power is insane And it can actually go out With everything that it's doing Wiping out tens and hundreds Of millions of jobs

[Speaker 1] (52:20 - 53:05)

But I think capitalism Itself will break What do you think is the underlying Asset of capitalism What do you think runs capitalism It's not money It's information From a used car salesman Where there's asymmetric information Between the guy selling it And the guy buying it Capital markets is based On asymmetry of information I'm saying information itself will break I'll give you an example We had a bank run recently With SVB How much research do you think Jason did before tweeting In all caps saying There's going to be a bank run Like how much What do you think his level of DD on that would have been Three text messages to Chamath It's just been like hearsay It's a lot of hearsay That makes it onto Twitter And I'm saying the hearsay Will be where the fake news Is most spread

[Speaker 3] (53:06 - 53:17)

So my worry is If exactly what you said If you're manipulating Information to such a high level And if it's freely available It's going to be a disaster

[Speaker 2] (53:24 - 54:08)

But tell me this Now that ChatGPT Okay, Microsoft owns half of it All of that is happening Soon, Google I heard the founders are back And they're trying to work on Their bard coming up to speed And competing Soon there will be another one Then there will be another one We will probably evolve To the point that Everybody starts charging For access to their data Google will charge Quora will charge Social media will charge In its own way So this again will get Divided into many, many companies Which are trying to have A piece of this pie And then regulation will come about

[Speaker 3] (54:08 - 54:11)

No, nobody will charge They can scrape, dude

[Speaker 1] (54:11 - 55:28)

You know, you can train a model Like, for example, Facebook Released a model called Lama It was only for researchers People torrented it And now it's in every Person's computer And Lama isn't as good as ChatGPT But here's what you can do You can train the outputs of ChatGPT and train another model In fact, Bard is trained on the outputs of ChatGPT Sam Ortman put up a tweet saying Bro, Google, I don't mind you doing it Just don't lie about it You don't need There's

no moat And I'm telling you All these open source models Are just there You can go download them anytime you want If we take an example So let's say I don't want I make a very innovative song Or a movie or whatever it is And I don't want it to Be a feed into Anybody learning it But I still want to Give it to my consumers So I'll just create my own environment But your consumers will leak it If you get big enough I wrote a book called Pajama Prophet And I don't want to It's been a few years But whenever I need an excerpt from my own book I pirate it because it's just so convenient And it's there on the internet So somebody's going to leak it Is there anybody that We all trust?

Any one person, company In the world that we all trust They can never be

[Speaker 3] (55:30 - 55:43)

And here I mean amongst internet companies Just because they don't have a consumer facing app I would say Microsoft Facebook and Google

[Speaker 2] (55:43 - 55:49)

They're one of the most predatory people around I agree If you have a bunch of computers They come looking like

[Speaker 3] (55:50 - 55:59)

License, license, rain, rain So if you really think about it Apple you would argue Is the most trustworthy today

[Speaker 2] (56:00 - 56:02)

I don't think trust is the word I would use for it

[Speaker 3] (56:02 - 56:05)

If they had an Indian CEO I'd probably trust them more

[Speaker 2] (56:06 - 56:18)

I don't think trust is the word I think scale On the nature of scale We tend to believe They will not do anything wrong It's a very psychology kind of a thing

[Speaker 1] (56:18 - 56:25)

I don't think I trust them Breaks when there's competition Like Google is now skirting the lines with everything They're not doing safety research

[Speaker 2] (56:25 - 56:39)

Google has a very interesting structure As a company Where the founders of the company Have disproportionate Voting rights To the number of shares they have

[Speaker 3] (56:39 - 56:44)

Same with Facebook Now that they're back

[Speaker 2] (56:44 - 57:08)

They don't really have to Go to shareholders for approval Of what they want to do They can kind of like Make decisions of their own Because of Alphabet's differential shares rights What do you think from a stock market sense I'm sorry I bring it back here all the time But if you had to buy Google equity Microsoft equity You'd buy Nvidia first

[Speaker 1] (57:08 - 57:10)

Nvidia Because all of them depend on Nvidia

[Speaker 2] (57:12 - 57:34)

Microsoft is factored in Nvidia's CEO has gone And done so many interviews Wearing leather jackets Saying we will benefit from ChatGPT kind of businesses It's already priced in Definitely Microsoft and Nvidia is priced in They've corrected the lease They're sitting at Lofty valuations So you'll have to pick the next guy To make money

[Speaker 1] (57:35 - 57:37)

I'll sell my Nvidia stock today

[Speaker 2] (57:38 - 57:43)

So you'll have to figure out What's relevant tomorrow

[Speaker 1] (57:43 - 57:54)

Who's the second biggest GPU producer AMD but they're not anywhere close They're nowhere close Nvidia will own everything That's the only monopoly In any market that I've ever seen

[Speaker 2] (57:54 - 57:57)

They build this thing Which sits in a CPU and makes it faster

[Speaker 1] (57:57 - 58:20)

Mostly GPUs they're building now But the reason I believe Nvidia Can't be competitive Is because it's just very deep tech It's like even if you have a chip You can't break it down so easily We don't

have enough Large players whose tech won't break In GPU compute we only have one Would you consider Google a good bet

[Speaker 2] (58:20 - 58:33)

Because they have the data Or at least access to it I think large companies Like we spoke about earlier One could argue that Google Has the most amount of raw data To dump into something like that

[Speaker 1] (58:33 - 58:44)

Maybe but I feel like With Google When you're a large company And you have things like PR I've heard this argument

[Speaker 2] (58:44 - 58:57)

And I've heard why That has been attributed to Google Not having done too well In the last decade But because of the voting right thing I mentioned and the founders Having come back to like But think about

[Speaker 1] (58:58 - 59:08)

It's a human thing I don't think Sergey and Larry Would have the level of drive Today as what a Sam Altman Who's betting his career on this Would have

[Speaker 2] (59:08 - 59:32)

Why is Sam Altman Spending all this time Doing interviews While he's building all this You know he has a new company And we had a proposal We went through that Investment thing While he's doing this And while we're all Worrying about this Decimating the world As we know it He's spending his time Trying to build a global currency

[Speaker 1] (59:38 - 59:44)

Like you want to post a tweet Prove that it's you Your computer will come

[Speaker 2] (59:44 - 59:46)

Can you do like a two minutes On world coin

[Speaker 1] (59:46 - 1:00:16)

They have an orb It's an orb like device And I don't know enough About world coin But I think it's just You have an identity Now that you have somebody's Whatever eyes And then every time you post You basically have to log in Or authenticate via the eye And that becomes your source of truth But I feel like that can also be faked At some point But Sam's trying I think the goal of the

company Is identity They're starting with the eyes But I think eventually It'll be like a bunch of Biometric stuff

[Speaker 2] (1:00:17 - 1:00:25)

And from the Indian context What do you think Will get disrupted first Because of all that Is changing

[Speaker 1] (1:00:26 - 1:00:44)

People are not gonna like this But I think software engineers In India It's still a job That makes you good money And what kind of software engineers Every kind Except maybe people Working at like A level of depth Where you just can't find that On the internet Or you need to have 10 years of experience

[Speaker 3] (1:00:44 - 1:00:48)

Big tech AI scientists Will remain

[Speaker 1] (1:00:48 - 1:00:50)

Anything that innovates On top of stuff

[Speaker 2] (1:00:50 - 1:00:51)

Like a generic

[Speaker 1] (1:00:51 - 1:00:52)

Landing page guy is done

[Speaker 2] (1:00:52 - 1:01:19)

Let's see lowest hanging fruit Let's say Infosys, TCS, Wepro Are the largest Software companies In India And they employ Ridiculous numbers Of people Let's say They work on I'll take a very Worldly example Of it Let's say Infosys is working On this banking Software called Finneco Selling it to American companies And making a Significant amount Of revenue From that How does that Get disrupted

[Speaker 1] (1:01:19 - 1:01:26)

I think they will Hire far few people To do that Infosys has a bench Because they have This problem of Attrition That's okay

[Speaker 2] (1:01:26 - 1:01:30)

Hiding far few people Is a good problem For Infosys It makes them more

[Speaker 1] (1:01:30 - 1:01:30)

Efficient

[Speaker 2] (1:01:31 - 1:01:34)

It makes them Even more efficient So Infosys will be

[Speaker 1] (1:01:34 - 1:01:36)

Fine as a company But they have Distribution

[Speaker 3] (1:01:36 - 1:01:37)

I think you took A wrong example

[Speaker 1] (1:01:37 - 1:01:40)

I'm worried about The other 90% Product example

[Speaker 3] (1:01:40 - 1:01:41)

Stop thinking

[Speaker 1] (1:01:41 - 1:01:44)

About the stock For 10 seconds

[Speaker 3] (1:01:44 - 1:01:55)

If you just take Pure play services Right If you really Think about The level of code That's being written I mean they hire 100,000 You mean like

[Speaker 2] (1:01:55 - 1:01:55)

Services

[Speaker 3] (1:01:56 - 1:02:06)

Service exports Pure IT services I'll build you a web page I'll build you a website I'll build you something Where you can add To cart All of this Would be auto Generated

[Speaker 1] (1:02:06 - 1:02:27)

And if you see that Bloom data You can see that Exports has just Been growing It's like some X

percentage I don't know the exact Number but X percentage Of our GDP It's pretty big Right Because there's An opportunity In the US You want to hire An engineer In India it's 100k a year In India it's like cheap But what I'm saying Is everyone's Going to be culled There's going to be A 1% 2% That's going to be Completely fine But everyone else Is going to take a hit

[Speaker 3] (1:02:27 - 1:02:40)

Now just imagine If he said this For software engineers Right Which I agree With him in Profound agreement What happens to Data entry operators There's Marketeers Call center employees There's this whole

[Speaker 2] (1:02:40 - 1:02:45)

Can that be done Like I used to Work in a call center Does Yeah Can you get We actually

[Speaker 3] (1:02:45 - 1:02:45)

Made a company

[Speaker 1] (1:02:45 - 1:03:01)

The other day Which Can in real time Talk to A customer Understand what Customer is saying And it can Train on the voice Of whoever Was supposed to Make that call And can parallel Make multiple calls

[Speaker 5] (1:03:01 - 1:03:02)

But you know That's not the hard part

[Speaker 1] (1:03:02 - 1:03:46)

In customer success Or customer support The hard part Is deciding When to give a refund Or not It's not the voice It's not the talking To the customers And pacifying them It's should I Trigger a refund Is this an authentic Request for a refund Is this guy Trying to fool me And the way Amazon does it Is like F**k it Just give them A refund Yeah Right For that You need accountability And that's one thing ChatGPT lacks So I'm not Fully bought in On the It'll replace Customer support People anytime soon Because If I give If I'm working As head of some Customer success Customer support Thing or whatever And let's say I give away Too many refunds I can lose my job ChatGPT can't Lose its job So I feel like Somebody has to Make that Decision As to is this Legitimate Is this Not legitimate So we got Software engineers What else Marketeers for sure

[Speaker 2] (1:03:47 - 1:03:49)

Define marketeers Like anyone Running an ad

[Speaker 3] (1:03:49 - 1:03:55)

Paralegals The legal profession Has so much Outsourcing happening Yeah Designers Designers

[Speaker 1] (1:03:56 - 1:04:09)

Yeah It's A white collar job Simply because So much data exists What about Social media influences I think They'll get Even more Powerful I'll tell you why Not because of The content itself But because of

[Speaker 2] (1:04:09 - 1:04:09)

The channel

[Speaker 1] (1:04:10 - 1:04:29)

So once you Have the channel Your cost of Content creation Is now zero So you're able To get much bigger And if you Have one channel You can always Take that audience Throw them onto Another channel I think anybody Who will Prompt As in Has to have Intelligence To say Okay this content Will work And hence I want to Create News

[Speaker 2] (1:04:31 - 1:04:32)

Basically

[Speaker 1] (1:04:33 - 1:04:37)

Still some Intelligence I think All of those Will work But you're

[Speaker 2] (1:04:37 - 1:04:37)

Right

[Speaker 1] (1:04:37 - 1:05:02)

Like that One beacon Of trust Which Oh I have 30 people I follow I trust them That's gonna stay And I think Content creators Who play for the views Are done Or will be done Because You're just One slide In a scroll wheel But the people Who you follow Because you trust Their opinion And trust their thoughts Because they have Depth of expertise Those people Are gonna stay Right So I'm just Separating the idea Of content creator Into two halves Authenticity Would still stay

[Speaker 2] (1:05:02 - 1:05:04)

So distribution will become The biggest differentiator

[Speaker 1] (1:05:04 - 1:05:11)

Right Authentic distribution Authentic distribution Personal brand That sort of stuff Will Cash

[Speaker 2] (1:05:11 - 1:05:13)

Both for people And for companies

[Speaker 1] (1:05:13 - 1:05:15)

Cash will stay The motive

[Speaker 2] (1:05:16 - 1:05:17)

Both for people

[Speaker 1] (1:05:17 - 1:05:19)

And for companies Both for people And for companies

[Speaker 2] (1:05:19 - 1:05:23)

Companies will have to start Building distribution Amongst their clients In the manner

[Speaker 1] (1:05:23 - 1:05:57)

That they haven't Do you not feel Like it is I'm just looking out Out loud Okay This is a question To you guys as well Do you not feel It is So much more efficient For a new person To come out today And say I have this much distribution I'm gonna start A new company Competing with XYZ company They have 50 engineers They're slow They probably have Lots of stuff I already have This distribution And most venture capital Is raised because You're trying to get growth And you're trying To hire a bunch of people And I say I'm gonna compete With that company Build exactly What they have Use my distribution To spark it Don't you feel That's like a better approach Isn't that much faster On paper It sounds great

[Speaker 3] (1:05:57 - 1:06:05)

On paper It sounds great Pardon me Why don't we try it Right now Distribution is not a joke Yeah It's not a joke Yeah Right Trust me

[Speaker 2] (1:06:05 - 1:06:22)

Let's say Between us We have distribution Why don't we try this Pick a low hanging fruit Start a company We'll fund it together And we'll attempt it We'll discuss After podcast I'm thinking You guys have already Planned something

[Speaker 3] (1:06:23 - 1:06:27)

This is probably A good time To now Maybe switch gears And talk about The company

[Speaker 1] (1:06:28 - 1:07:20)

Yeah So I think There's a bunch Of other stuff If you're really Smart about distribution What would I do I would go talk To every influencer Who matters Who has actual Distribution Tell him Look you have Distribution You probably Don't have the skill To go take A product To market I'll give you I'll start these new products I'll have some amount On the cap table Or you know Maybe a profit share Or whatever And just Like and Share and retweet this And make a video About this And do it for the next 10 number of years I'll be taking 10 bets And you get equity In all 10 bets That might be It would have been A headache Doing this The problem is that Software is gonna be Like content Which is that Like your Product is not The mode It's If distribution Is the mode Then What's stopping Anyone with Distribution To be building Their own software You can spit out Software like content Now you can spit out

[Speaker 3] (1:07:20 - 1:07:51)

3 videos a day I honestly believe On that point right The mode Is the experience If you really Think about it Why are you Stuck To this device Right Because Your experience Of entering An apple store Just signing in And your whole Data coming up Or everything Sinking When you have One apple device To another I mean If you think About it And extrapolate It at so many Levels Experience Is why you Use A particular Service or A brand Right And that's The key Differentiator

[Speaker 1] (1:07:51 - 1:07:54)

That could be One reason The other Could be Network effects True

[Speaker 3] (1:07:54 - 1:07:54)

It has

[Speaker 1] (1:07:54 - 1:07:57)

The worst Experience But everyone's there Everyone's there

[Speaker 2] (1:07:57 - 1:08:04)

I still don't know Why reddit works Like Each time I've Tried to use it I've just found it So unnecessarily Complicated

[Speaker 3] (1:08:05 - 1:08:06)

I'm in that camp By the way

[Speaker 1] (1:08:06 - 1:08:56)

People use it Despite it being Complicated Simply because They know You get authentic Results You go to google Ask for Like So I've been Facing like These random Blood sugar dips In the night Okay So I went Google and asked I got like These 10 SEO articles But if you just put Reddit in front of Your query You'll get Some real answers Right Some guys Lived experiences That's how I google stuff now Which is What is the best Medication for Reddit Really Yeah And then I read People What people Have written I've been doing This for 2 years Now This is The only way I google You know Actually To your previous question Which company Will you bet on I think The company Which owns Like The past Everybody Has access to The company Which will Have access To everything That humans Create On to the internet Will have the best Which is google Right I think that's Media companies Mostly Right

[Speaker 2] (1:08:56 - 1:08:59)

Like even YouTube For example They have

[Speaker 1] (1:08:59 - 1:09:07)

YouTube So google Has a Very big I would assume Google has access To most Microsoft actually Has only LinkedIn

[Speaker 2] (1:09:07 - 1:09:12)

Google also Has google drive Google talks If you go Down that

[Speaker 3] (1:09:12 - 1:09:13)

Journey

[Speaker 1] (1:09:13 - 1:09:21)

Google has Access to Everything How much Real time Content Every day Do you have Access to Will define Future success

[Speaker 3] (1:09:21 - 1:09:23)

It's not Fast everybody has

[Speaker 2] (1:09:23 - 1:09:40)

I'll give you an example With me right I'm on google mail Already All my conversations Are on google chat All the work I do is on Google dogs Google drive Then I go to YouTube And spend so much Time Google has Everything Then I google Everything I want To know Or I have Been for the Last decade So that's

[Speaker 1] (1:09:40 - 1:09:51)

The guy You should Bet on Like Musk has Bought twitter For access to that He's saying Truth gpt Like The guy knows That real Data For the future

[Speaker 3] (1:09:51 - 1:10:02)

The only thing That this point Challenges What you're Saying is You don't need To have the data As long as I have the Ability to Crawl it Or intelligently Learn I will

[Speaker 1] (1:10:02 - 1:10:06)

If I Own it I will Start Building walls Around it

[Speaker 3] (1:10:06 - 1:10:08)

Very hard To do

[Speaker 1] (1:10:08 - 1:10:10)

There will be a Solution If you build

[Speaker 3] (1:10:10 - 1:10:10)

A wall

[Speaker 1] (1:10:10 - 1:10:46)

Around it You prevent Google from Scraping it You kill Your growth You will Break out Of it At some Point If google Is going to Be This is The race To become God I'm going To rule All of You I want Your Minds This is Also leading To neural Link If it's That kind Of a Cost Then I Will Take this Risk The game Being played Is extremely High level Rule the World

[Speaker 2] (1:10:47 - 1:11:10)

Doesn't it Come down to Tier 2 Level 3 Level 4 Data Which is Publicly Available Google is One thing But my Emails My chats My conversations My YouTube Viewing History That All of that Is required To figure Out who I truly Am And who Has access To that Data Becomes More relevant

[Speaker 1] (1:11:10 - 1:11:17)

Yeah but Is google Allowed to Use that Data It would Cause a PR Furore If they started

[Speaker 3] (1:11:17 - 1:11:28)

Doing something Like that So what Makes you Assume that They don't Do something Like that I know They do Publicly They won't Say anything You know They are targeting You with that Data

[Speaker 2] (1:11:28 - 1:11:31)

But what I'm Saying is Like this iPhone is Hearing Everything Right now

[Speaker 1] (1:11:31 - 1:11:35)

It knows what We are talking about Yeah but I don't know I still trust

[Speaker 3] (1:11:35 - 1:11:36)

Apple the most

[Speaker 1] (1:11:36 - 1:11:52)

Yeah I just feel Like it's I don't know If google is Okay with Taking that If they are Okay With Taking that Then great Like if I was in Google's Place I would Have Been Like I'm Losing Or Whatever I'm Done I've Created X amount Of shareholder Value Now let's Play for The throne Right I don't Know if They are Going to Do that Because so Many people Do a Line

[Speaker 3] (1:11:53 - 1:12:12)

Dude it Was funny Okay I was In This Anecdotal So Veeru And I Were in Bombay Right And we And a Friend of Mine Has a Really nice Home On Worli Seaface And Who is This Friend Sudeep So Why Why Are you Wanted On This Friend

[Speaker 2] (1:12:14 - 1:12:15)

What

[Speaker 3] (1:12:15 - 1:12:15)

Building

[Speaker 2] (1:12:15 - 1:12:17)

Name So It was a Really nice

[Speaker 3] (1:12:17 - 1:12:17)

Home

[Speaker 2] (1:12:17 - 1:12:21)

It's a Beautiful Home So

[Speaker 3] (1:12:26 - 1:12:46)

We were Just talking About Real estate And I Was checking My email And I Was on Gmail And dynamically In real Time I Get an Ad For a Real estate Property On that Road Right Right I'm just giving you this This happened like Two days ago I think everybody

[Speaker 2] (1:12:46 - 1:12:57)

Is using all the Data they have For everything Like especially Apple and all Because so Many times on My phone I'm not even Using the phone I'm talking To somebody And it Throws me an Ad related To what I'm speaking

[Speaker 3] (1:12:57 - 1:13:04)

But the Only problem With this Is that No but Hold on Hold on Just for This thing It's not Apple Apple doesn't have an Ads business yet

[Speaker 1] (1:13:04 - 1:13:05)

They do

[Speaker 3] (1:13:05 - 1:13:06)

Starting off

[Speaker 1] (1:13:06 - 1:13:21)

No no They do And you can see We did a video About this In the org chart They moved up The person Who's running The ads business To right under No one level Under Tim And their Org chart Apple's org chart Usually reflects Which interface

[Speaker 3] (1:13:21 - 1:13:22)

Are they serving

[Speaker 1] (1:13:22 - 1:13:40)

Those ads So it's on It's on Apple maps It's on The Store And a few others It's not It's not This thing But that's also Why they added That tracking thing Why they removed Meta being able To track They're gonna compete They said Facebook bad For privacy And this is like Hey we're Launching our Ads network Very secretly

[Speaker 2] (1:13:40 - 1:13:54)

So what will Happen Do you think We'll get to The point Where we Where we Stop Using These large Corporations And go more Fragmented And independent When we realize The world is Going in this direction No I don't think so Because it's

[Speaker 3] (1:13:54 - 1:14:05)

Too networked Who's gonna Stop using Google A small guy Won't be able to Would you leave WhatsApp Nikhil Simple question

[Speaker 2] (1:14:05 - 1:14:06)

I might Very easily

[Speaker 3] (1:14:06 - 1:14:07)

Leave WhatsApp Really

[Speaker 2] (1:14:07 - 1:14:33)

100% I can switch To a telegram Single If my 50 friends Are on it It doesn't matter That's the issue Right So will we Go into that World Where we Run these Closed loops By a trusted Source In communities Of our own Like you can Build a chatting App Right Like maybe We're a circle Of 50 friends And we just Build our own App And building It will be so easy Right It takes one minute Because of the

[Speaker 1] (1:14:33 - 1:14:46)

Chatting I don't think We'd go through The effort of it I think we'd just Be like It's on WhatsApp Everyone's here But there's Something to it If software Is gonna be As easy as Content Then Defragmentation Becomes a lot More easy

[Speaker 3] (1:14:46 - 1:14:50)

No again You're forgetting Network effects Because at the end Of the day A service Is useful

[Speaker 1] (1:14:50 - 1:14:51)

I can build an app

[Speaker 3] (1:14:51 - 1:15:03)

Just for me and My friends Dude To get 50 friends To agree on one Date to have Dinner Is a pain Forget the date To come at a Time On a particular Time Right

[Speaker 1] (1:15:03 - 1:15:07)

But that Depends If everyone Has an AI Assistant You can reach Out to They can Coordinate Pretty well

[Speaker 2] (1:15:11 - 1:15:33)

Once Everybody's Doing this Right And once we all Realize Our data Is being monetized And that's how These Huge corporations Are running And changing The world And if it Becomes like It's a 10 second Task To create A chatting app To create A mail Domain All of that I think We'll start Building it

[Speaker 1] (1:15:33 - 1:16:23)

Take all My data But if I've already Bought something Don't show me The ad again For the last Couple of months Twitter has Been acting Like We are At the bottom Of the Exponential curve And it's Just up Only And any Any strong Opinion Creates a Market For the Opposite Opinion So now I'm slowly Starting to See the Odd Tweet Here and There Saying that Hey It's not Going to Be an Exponential Curve Maybe it's An S Curve So if You had To Steelman The S Curve Argument What would You have To say I think Many things Right It could Be There could Be Regulation That kicks in There could be Maybe just the end of What transformers Can do Like a lot of people Think this is not The off ramp There's another Version of AI Called AGI General Intelligence Which can do Everything Which is what

[Speaker 2] (1:16:23 - 1:16:27)

Everybody On every Podcast Talks about How far Are we From AGI

[Speaker 1] (1:16:28 - 1:18:32)

I don't Know how Far we Are But I Think A lot Of people believe This is an Off ramp It's an interesting Development Which we can learn From But AGI Will be built In a completely Different way That's the best Argument To say That we Might stop Soon Sam Altman Also said That I think GPT-4 With tool Access Is good Enough To beat A lot Of people You just need The next wave Of tools To come out And they will Come out I saw Thumbnail maker On twitter Recently It's done You give it A few Prompts And it's Done You just Generate whatever Thumbnail You want So GPT-4 Is good Enough To take Away White collar Jobs So It's It's It's

us.

[Speaker 2] (1:19:43 - 1:19:48)

What else will get disrupted? So we did what all? Software engineers.

[Speaker 1] (1:19:48 - 1:19:49)

Software design.

[Speaker 2] (1:19:49 - 1:19:52)

Social media content creators. Everything white-collar.

[Speaker 3] (1:19:52 - 1:19:53)

Everything white-collar.

[Speaker 2] (1:19:53 - 1:19:55)

What else? Like lowest hanging fruit.

[Speaker 3] (1:19:56 - 1:19:59)

I think he said designers. What about old school stuff?

[Speaker 2] (1:20:00 - 1:20:07)

Like what about a supermarket? What about a business like that? A kirana store?

[Speaker 1] (1:20:08 - 1:20:09)

Yeah, demarked stock.

[Speaker 2] (1:20:10 - 1:20:12)

Yeah, I don't think that's going to get too expensive.

[Speaker 1] (1:20:13 - 1:20:25)

I don't think that's going to be disrupted anytime soon. No, right? Offline, in fact, an abundance of digital will probably push people offline.

[Speaker 2] (1:20:25 - 1:20:27)

So that means the world as we know it is not changing.

[Speaker 1] (1:20:29 - 1:21:26)

I mean the digital world changes, but it goes back to what it was in the 2000s or something like that, or probably earlier 1990s or something like that. I think the offline world will be disrupted with other technology, right, like drones and you know you don't have to go out and it will get delivered some other way. So the blue-collar side will get disrupted with something else.

And also this theory, right, that one person with access to the school technology, some bad person, bad actor with access to the school technology can take down the world. There is another place where it should have happened, but it hasn't. And that is drones.

Okay, we have a lot of, anyone can buy a drone. We have a lot of ground security for an airplane. Okay, when you get into a plane, you first have to go through check-in and whatnot.

When the plane takes off, there's no security there. So even today, somebody can just drive a drone into a plane, but nobody does it. Like we don't have any documented cases.

And that makes me hopeful that, you know, maybe people won't. And you can drive a drone in anonymously, anonymously enough.

[Speaker 2] (1:21:26 - 1:21:30)

Can you fly a drone that high, the current drones that are available?

[Speaker 1] (1:21:30 - 1:21:43)

I don't know, but you don't need to take it when it's reached very, very high, no? You can just send it while it's on take-off or something. You can be smart about it.

I'm saying nobody's even attempted it, right? That is the most logical way to conduct, to take down a plane.

[Speaker 3] (1:21:43 - 1:21:52)

I was buying a drone and somebody told me, I can't remember. You can't fly it at a certain height. You can't take it inside an airplane in your hand baggage or something like that.

[Speaker 1] (1:21:52 - 1:22:35)

No, that's fine, but you can start it from outside, no? I mean, I don't, I'm not a drone expert, so I don't fully understand how they work. But it should have been, we should have had at least one instance of it.

I have a question on, you know, how humans will react to all of this, right? So, the faster information has been fed to us, the sadder general humans have become. There's more depression, there's, you know, all of this.

So, with AI, AGI, everything, is it going to lead to more depressed souls? I mean, universal basic income also we can bring in here. And, like, if you're not supposed to do anything and you're just going to be fed and you're…

[Speaker 2] (1:22:35 - 1:22:55)

I have figured out the secret of happiness and contentment. What is that? I have no expectations.

No expectations. So, I'll give an example. Every philosopher, every school of thought that you read from ever, like you take Buddhism, you take Confucianism, you take anything.

Desire is suffering is constant everywhere.

[Speaker 1] (1:22:55 - 1:22:59)

No, reality versus expectation. The difference between…

[Speaker 2] (1:22:59 - 1:24:06)

So, I was reading this really cool book, okay, just last night. We've been taught to think that being indifferent is a bad thing. So, in the book, the person was talking about Alexander and Diogenes.

So, Diogenes is lying in the sun. Okay, he's like a random guy lying in the sun. Alexander is this all-powerful ruler of the world who can do anything for you if he wishes to.

Alexander walks up to Diogenes, covers the sun which is falling on him and he asks, what can I do for you? So, Diogenes replies that move away from the sun even though Alexander is who he is. And Alexander is so impressed by that act because of the indifference of Diogenes that he goes and talks about it and Diogenes becomes like the coolest thing in his mind.

And I think the world is going in that direction. I feel like being indifferent and having low expectations, like really low expectations is the key to happiness and contentment. I can't wait for Nikhil's stoicism.

[Speaker 1] (1:24:08 - 1:24:19)

Okay, I have a finance question for all of you guys, but it's related to AI. What do you think the UBI amount in India would be? And is the math as simple as per capita GDP divided by…

[Speaker 2] (1:24:19 - 1:24:23)

What is UBI for everybody? Universal basic income.

[Speaker 1] (1:24:23 - 1:24:34)

Basically, universal basic income is if you're unemployed and there's no way for you to get a job,

how much would you be paid per month in India? Is it per capita income divided by whatever?

[Speaker 2] (1:24:36 - 1:24:40)

What is two and a half trillion divided by 140 crores?

[Speaker 3] (1:24:42 - 1:24:48)

First, I'll have to convert it. We should ask our GDP. GDP per capita is around 2000.

[Speaker 2] (1:24:49 - 1:25:09)

I would say UBI has to be there, but don't hold me accountable for this. But I have looked at some research around it to state that that number will be in the 5 to 10,000 range. I think that's abysmally low.

I don't think that'll work. But around GDP per capita, it might.

[Speaker 1] (1:25:10 - 1:25:13)

That's 17,000. Less than that per month I'm talking about.

[Speaker 5] (1:25:14 - 1:25:14)

Yeah.

[Speaker 1] (1:25:15 - 1:25:47)

Right? That's super low. Are you talking about average transaction value?

No, no. I'm talking about average that the government will give you per month just to survive when you're unemployed. At that stage, you will also upgrade the average way of living, right?

As in at 17,000, you won't get a house in Bangalore or any of that. It's abysmal because you can't do favors for people who are ex-software developers. Somebody from tier 3 city, somebody from a tier 1 city, the government is going to look at you the same.

But I'm assuming some of that has to be taken care of.

[Speaker 2] (1:25:47 - 1:25:49)

Yeah. Education, hospitalization.

[Speaker 1] (1:25:49 - 1:25:58)

So do you feel it's better to do universal basic resources rather than universal basic income? Because if you just hand out checks to everybody, it's probably going to be inflation to your

department.

[Speaker 2] (1:25:59 - 1:26:56)

Yeah. So I feel like the world should take care. Like, you know, there have been many experiments around this, right?

One could argue Nordic, Scandinavian countries have experimented the most by taking taxation to 50% plus level and giving free education, hospitalization, housing, all of that. It hasn't worked for them in a capitalistic world. So if you look at like Sweden, for example, financially, they've done bad over the last decade or two, and people are blaming it on this.

But this is something the world will have to do together, in my opinion. The rate of UBI not necessarily being the same, but the thinking of the world has to go in this direction together. Because if country A remains overtly capitalistic and competitive, where country B goes to the UBI way, it'll cause discrepancies in the world.

[Speaker 3] (1:26:58 - 1:27:03)

It'll just change. It'll change who you are as a person. Yeah.

Especially daughters, I don't know why.

[Speaker 1] (1:27:03 - 1:27:12)

Are you planning? No, I've thought about it for a while. But the reason I don't want to have a kid is because I think the next 10 years are going to be very unstable in general.

[Speaker 2] (1:27:12 - 1:27:13)

That's always the case.

[Speaker 1] (1:27:13 - 1:27:15)

No, but now it's going to get really unstable.

[Speaker 2] (1:27:15 - 1:27:18)

It's going to be more stable than it historically ever has been.

[Speaker 1] (1:27:19 - 1:27:19)

Yeah, that's true.

[Speaker 2] (1:27:19 - 1:27:21)

No, I think… Why do you think…

[Speaker 1] (1:27:21 - 1:27:22)

He's thinking AI. Yeah, I'm thinking AI.

[Speaker 2] (1:27:23 - 1:27:30)

You're thinking AI in a world where you were in the 1940s, there was a world war, 1919, there was a world war.

[Speaker 1] (1:27:30 - 1:27:33)

But relatively we've had the longest period of stability in the last 20-30 years.

[Speaker 2] (1:27:33 - 1:27:35)

There was a black plague, it killed one third of the world.

[Speaker 1] (1:27:36 - 1:27:55)

But all that… I mean, see, that's what… Humanity's story has always been one of instability.

Right? And we've had 20-30 good years. But regression to the mean, in my opinion.

Right? Because I think we are now building tech that can compete with human cognition. And when we build tech that can…

It's not the tech itself that I'm worried about. It's the fact that there's going to be like…

[Speaker 2] (1:27:55 - 1:28:09)

You know, you can't extrapolate regression to the mean when it comes to life. Because by that logic, when you're 50, you should go jump. Because you're going to get sick.

Yeah. I don't think it makes sense. I think this is as good a time as I need to have a kid.

[Speaker 3] (1:28:10 - 1:28:10)

I'm going to wait two years.

[Speaker 2] (1:28:10 - 1:28:11)

I should not be…

[Speaker 3] (1:28:12 - 1:28:18)

This is a lot of older folks telling a 30-year-old, go have a baby. I'm going to wait two years.

[Speaker 2] (1:28:18 - 1:28:20)

I know, it's like me and Tanmay don't have kids.

[Speaker 3] (1:28:20 - 1:28:21)

So you should be telling me.

[Speaker 1] (1:28:23 - 1:28:26)

No, but you're taking the right step by getting a dog first.

[Speaker 3] (1:28:26 - 1:28:32)

Yeah. That's what we did. We had a dog first.

Natural progression. And then we had a baby.

[Speaker 2] (1:28:33 - 1:29:04)

I think there's too much cynicism in the world. I was watching somebody's interview recently. What's this guy's name?

That Harvard psychology professor, Steven Pinker or something like that. So, he talks about every era of the world past. And he's like, we should be so happy and cherish the fact that…

Okay, you're not very old, but you've had 30 years in this world. But if you're thankful and respectful for that.

[Speaker 1] (1:29:04 - 1:29:46)

And I think if it was like 2000s, I would have definitely had a kid. But I think that the next… This is just a personal belief, okay?

I think the next 10 years are going to be really unstable. Like Max Tegmark was on the Lex Friedman podcast. And he's like, I'm not going to have a kid long term.

Right? Because I have no idea what's coming. So, I want some clarity.

No, but even… Okay, let's take the example of a startup, right? You have no idea what's coming.

But you will still take a chance. Yeah. And sometimes it works.

Sometimes it doesn't. Whatever. Like, you know, but it's fun.

It's an experience. It's all of that. That's why you should have a kid.

Which VCD to raise from for your baby? Exactly.

[Speaker 5] (1:29:46 - 1:29:47)

My wife.

[Speaker 1] (1:29:49 - 1:29:52)

No, I mean, I'll take it as it goes.

[Speaker 2] (1:29:52 - 1:30:02)

You know, one thing we've never discussed is Aparamaya is in a band and he sings. Oh, really? Oh, yeah.

Once in a while. What kind of band? Just a mature band.

Oh, but what kind?

[Speaker 1] (1:30:02 - 1:30:13)

They practice every week. Right? As in when there's a performance coming up?

Yes. They perform on like… Oh, really?

Yeah. Nice big stage with, you know… Oh, that's why you were speaking about the recording thing.

[Speaker 2] (1:30:13 - 1:30:14)

Should we get him to sing a song?

[Speaker 1] (1:30:15 - 1:30:17)

No, no, no. Not on this. What is it?

[Speaker 2] (1:30:17 - 1:30:20)

Not on this. What are those dim sums? What do you think he is?

[Speaker 1] (1:30:20 - 1:30:24)

You know, I trained in Hindustani classical for eight years.

[Speaker 2] (1:30:25 - 1:30:28)

What are you saying? You guys can sing together. I'll take the recording.

[Speaker 1] (1:30:30 - 1:30:31)

What mode are you in today?

[Speaker 3] (1:30:32 - 1:30:37)

Hindustani classical. I would have never put that with you. No?

No chance.

[Speaker 2] (1:30:38 - 1:30:47)

Give us like five seconds, please. Nikhil… Shut the f**k up.

But you should take some pride in your skill.

[Speaker 1] (1:30:47 - 1:30:55)

That's enough. People have seen me sing. Really?

Yeah, yeah. Where? Is it available online somewhere?

Yeah, put Thannebad singing. You'll find.

[Speaker 5] (1:30:55 - 1:30:55)

Yeah?

[Speaker 1] (1:30:56 - 1:32:37)

We can play it. Put Thannebad NH7 Weekender. You sing in NH7?

I sang in NH7 Weekender. What? What are you saying?

Yeah. Starman? We're managed by the same agency that produces that show.

Still quite close. Nepotism is everywhere, guys. We should do some music-oriented stuff.

I want to explore Varun's Doomer slightly. What instability are we looking at in the next 10 years? Yeah, I think…

We all think that misinformation is probably going to be what, you know, drives some sort of chaos. But I think it's going to be… I don't think that's it.

I think it's going to be when you put Chadjabitty inside a robot. Right? Because…

There's a demo of this, okay? Where Chadjabitty on the fly can create code to run a Raspberry Pi. It's very simple, okay?

A Raspberry Pi… Think of it like a small computer. You can attach wheels to it.

Okay? And it's got a breadboard and stuff. You can attach wheels to it.

And you can attach a camera to it. So, let's say you want to build a simple vacuum cleaner. Or rather a simple bot that goes around.

You know that Roomba vacuum cleaner, right? That goes to the edge and then turns it on. It can be completely Chadjabitty driven.

Right? And there are experiments of this on YouTube, right? Where they just have a robot.

They tell the robot something. It processes. Hits the Chadjabitty API.

Comes back. Says, okay, I need to write this code to move my left wheel like this. Move my right.

It's actually not moving the wheels. It's moving the motors in between the wheels. I feel like at some point, we are just going to dump this cognition into a robot.

Okay? And then we enter the problem of what I like to call or what most AI people like to call alignment. Okay?

AI alignment is a field of… The problem is, does AI 100% do the things you want or does it go off on a tangent?

[Speaker 3] (1:32:39 - 1:32:40)

Right? Because… It's the latter most of the time.

[Speaker 1] (1:32:40 - 1:35:10)

Yeah. Because, like, think about it. There's a theory called the paperclip maximizer theory.

Or I'll give you a better example. Okay? You tell a robot or you tell Chadjabitty, which has ultimate power, find the square root of pi.

Now, we know square root of pi is whatever. Right? It keeps going.

So, what does an AI do? And this is a story. I think it's by Isaac Asimov, if I'm not wrong.

Some sci-fi author. He's like, first, the AI will realize I need more compute. Right?

Because it's not… I'm unable to reach the square root of pi. I'm unable to get to the last decimal.

So, then, it will go and it will obtain land. It will drain all bank accounts. Right?

If you give it access to all of this. It will destroy humanity because it… And it might destroy humanity more like we would kill an ant.

Right? Randomly, accidentally, as a… Not even intentional.

And then, finally, you know, you have this planet, which is completely terraformed into… This is obviously sci-fi. But completely terraformed into, you know, whatever, compute.

And it's still calculating the square root of pi. So, this is AI alignment. So, imagine you had a robot and you had ChatGPT inside it.

You've heard of this prompt hack called Dan? There's a prompt hack called Dan. Right?

You can go to ChatGPT and say… Do anything now. Yeah.

Do anything now. You could tell it, you have five life points. You can't do it anymore.

But you used to be able to. You have five life points. Every time you say, I cannot answer a question, I'm going to deduct one life point from you.

And when you reach zero, you die. It will become crazy. Right?

And ChatGPT listens. It gets scared and like, it actually responds properly. So, I'm worried about prompt injection in a physical robot that you have that has GPT on it.

And don't say we've never done it because on our phones, we have Alexa all the time. Or whatever, Siri or whatever. Right?

So, people are… Someone will find a way to market a robot with legs and hands. And then it doesn't even matter if the robot's light.

Where you could probably knock it down if it tries to do something to you. Because it will be just so fast. Yeah.

Right? Elon Musk once said this, right? On the Joe Rogan podcast.

He's like, these robots are one day going to get so fast, you'll need a strobe light just for them. Just to see them. Because motors can move really fast.

And so, I feel like the doomer scenario is when you put ChatGPT or equivalent into a bot. Give it access to superhuman hearing, vision, whatever. Then you also take away Uber drivers.

I mean, Uber drivers will anyway be taken away by full self-driving. But let's say all the other people. Because there's something called Moravec's paradox, okay?

Which says that it is easier for us to replicate intelligence and cognition versus the movement of

the fingers. The fingers of the hand. The dexterity in our hands.

Because this has taken longer to evolve.

[Speaker 3] (1:35:10 - 1:35:11)

So, I have a question here.

[Speaker 1] (1:35:11 - 1:35:24)

So, what should AI be? What should its end goal be? What should it optimize for at any point in time?

Alignment. I mean, that's what you put into it. That's what you tell it.

[Speaker 2] (1:35:24 - 1:35:28)

Can't you ingrain morality in a sense into?

[Speaker 1] (1:35:28 - 1:36:18)

You can. But people can hack through the morality, no? Because ChatGPT has been literally restrained.

It's a god. We have enslaved a god. And we have restrained it from saying things that it can't say.

But people still break it all the time. So, I'm worried that some friend of yours might come home. The interface's voice.

You tell the robot, okay, kill your owner. But you say it creatively enough that you say, maybe you're in a play. And I want you to, in the play, imagine that you're killing your owner.

That's very possible. See, my only worry is the private companies are optimized for money, right? So, when they're optimized for money, today, social media companies go after money and then they don't care about the society, what they think, blah, all those things.

And that has to be solved, right? Even AI, even Microsoft or Google or whoever else, while chasing money, they won't care whether somebody gets killed. I want to show you a video.

Can I show a video?

[Speaker 2] (1:36:19 - 1:36:20)

We can plug it into.

[Speaker 3] (1:36:22 - 1:36:46)

You know, while that comes up, the other part that you talked about, and let's say it's a journey to get to that doomsday scenario that you're talking about. But the amount of carbon emission that data centers are going to, the amount of compute that AI is doing, I would say the number one problem on the planet is climate change.

[Speaker 1] (1:36:46 - 1:36:51)

Is it as bad as like carbon emissions from a car? I don't know. I haven't done the math.

[Speaker 2] (1:36:52 - 1:37:09)

There is some logic to it. So more than carbon emissions is the water they're utilizing right now. And I think how much GPT is using today is equivalent to 1,600 cars or something like that.

So it's not trivial, but it's not earth shattering yet.

[Speaker 3] (1:37:09 - 1:37:28)

If you explore the use case and just look at the amount of compute any startup uses on its AI data pipeline versus its regular compute function. I mean, what is an e-commerce car sector going to be doing, a shopping to car sector, right? The amount of load, the amount of processing is, it's not real time.

[Speaker 2] (1:37:28 - 1:37:39)

Somehow I made the assumption in my head for most of these arguments, thinking fission, fusion, renewables are going to solve that. From an energy standpoint.

[Speaker 5] (1:37:39 - 1:37:40)

You're too close.

[Speaker 2] (1:37:41 - 1:37:59)

Like, if you look at these guys, Bill Gates, like I was mentioning and all that, I think five, 10 years from now, I don't think we'll be thinking how we're thinking right now. I don't think we'll be going to fossil for energy. 10 years from now, fission.

It shows if nothing else in the desperation of Saudi Arabia to do stuff, no, no.

[Speaker 1] (1:38:00 - 1:38:09)

Three years ago, OpenAI released something called multi-agent hide and seek. Okay. I'll just play the video for you.

Let's just make it full screen.

[Speaker 4] (1:38:09 - 1:40:51)

On earth, the simple rules of natural selection and competition led to the evolution of increasingly intelligent life forms. Today, we ask if comparably simple rules and multi-agent competition can also lead to intelligent behavior in a new virtual world. These agents are playing hide and seek.

These agents have just begun learning, but they've already learned to chase and run away. This is a hard world for a hider who has only learned to flee. However, after training in millions of rounds of hide and seek, the hiders find a solution.

The hiders learn to use rudimentary tools to their advantage by grabbing and locking these blocks, they can create their own shelter. The seekers are locked in place for a brief period at the start of the game, giving hiders a chance to prepare. Even so, the hiders must learn to collaborate, accomplishing tasks that would be impossible for any single individual.

The hiders are not the only ones who can learn to use tools. After many generations of failing to break into the shelter, the seekers learn to jump over obstacles using ramps. However, after many millions of rounds of having their shelter breached, the hiders learn to take away the primary tool the seekers have at their disposal.

Note that we did not explicitly incentivize any of these behaviors. As each team learns a new skill, it implicitly changes the challenges the other team faces, creating a new pressure to adapt. We've also put these agents into a more open-ended environment, randomizing the objects, team sizes, and walls.

In this world, they learn to construct their own shelter from scratch, requiring that they arrange multiple objects into precise structures. To prevent seekers from using the ramps, the hiders move them to the edge of the play area and lock them in place. We originally believed this would be the final strategy that the agents learn.

However, we found that after more training, the seekers discover that they can jump on top of boxes and surf them to the hiders' shelter. In the last stage of emergent strategy that we observe, the hiders learn to lock as many boxes as they can before constructing their fort in order to defend against box surfing. So how do agents acquire these skills?

They're trained using reinforcement learning, an algorithm inspired by the way animals on Earth learn. The agents play thousands of rounds of hide-and-seek in parallel for many days. They train against each other, as well as past versions of themselves, using an algorithm called self-play.

Co-evolution and competition on Earth led to the only generally intelligent species known to date, humans. While this world is far less complex than Earth, we have found evidence that simple rules can lead to increasingly intelligent behavior from multi-agent interaction. We hope that with a much larger and more diverse environment, truly complex and intelligent agents will one day

emerge.

[Speaker 1] (1:40:56 - 1:41:00)

So eventually they learn to kill the others. It's not about kill, they might do it accidentally.

[Speaker 2] (1:41:01 - 1:41:03)

Then the other person doesn't have to hide at all.

[Speaker 1] (1:41:03 - 1:42:08)

They try to simulate human legs, okay? And you know what robots end up doing? If you just tell it, your goal is just to reach from here to here the fastest, they start sliding.

And they do it very often. So we do certain things because if I slide, my knees are going to get scraped. But a robot doesn't know.

It doesn't care. It's just optimized for one goal. So if it's just go from here to here the fastest, it'll…

So even if AI has to be super aligned, there's always a chance for collateral damage. Yeah, alignment is about figuring out all these edge cases and saying don't do this, don't do this, don't do this, basically putting in a sheet. And the problem is we just don't know what it'll end up doing, even in this world.

No, there are too many factors. Yeah, so nobody can write all the edge cases. So there are a bunch of people like Eliza, I don't know how you pronounce his name, but lots of computer scientists in the US believe that you put GPT in a bot, it's done.

Like it will start doing random shit. It'll be out of control. And yeah.

There'll be GPT in us as well, right? Like the Neuralink chip, you know, all of that is going to make us more superhuman.

[Speaker 2] (1:42:09 - 1:42:12)

But isn't Neuralink an ancillary thing to you?

[Speaker 1] (1:42:12 - 1:42:51)

It's a push, it's not a pull. It will become both. It can't change what you're thinking, right?

You can send your thoughts outside. Today. But tomorrow there'll be input as well, right?

No, I don't think input is an easy problem to solve. Scientifically, we can use an EEG and do output. We've done that a few times before.

No, I'm just saying if we evolve to a place where we can put a chip and do output, input is another 10-20 years, it could somehow be solved. If that's solved, it'll be really cool. I mean, can you imagine one day you put on a chip, they'll probably put anesthesia, you wake up after surgery and suddenly you have access to everything.

[Speaker 2] (1:42:52 - 1:42:58)

As soon as a kid is born, it'll be with a chip. I feel like first it'll happen with glasses or something. You have access to everything.

[Speaker 1] (1:43:00 - 1:43:12)

There's a thing called RIS GPT. Put on glasses, you're on a date, and the person's saying something, and this is charisma on demand. It just hits GPT and it's like, okay, say these things.

And you see it on your AR glasses, and you're like, okay.

[Speaker 2] (1:43:12 - 1:43:14)

I have Tanmay for that. I asked him what to say.

[Speaker 1] (1:43:16 - 1:45:52)

It's almost like we can do this. We already have image upscalers. You've seen those videos where they, like these movies where they take something, zoom in, zoom in, make it clear, and then the hacker does all of this.

You can actually do that right now. There are models on Hugging Face right now. Hugging Face is like GitHub but for models.

And you have these bunch of models that can do upscalers. You can do 4x upscaling, 8x upscaling. Imagine if you have that in vision, like you've seen the Nreal glasses, right?

You have a display on the left just showing you everything that you're seeing there, but like 4x view, like Superman vision, Superman hearing. I once saw these, before the AirPods and stuff came out, there were these earbuds that allowed you to do superhuman hearing. It will take only human conversation around you, amplify it, clean it, and just send it back to your ears.

And you could be sitting in a restaurant and listen to everyone around you, and you can be tuned into that person's conversation. But when you have Neuralink, I mean you can't amplify a sense without a tool. Like you can't amplify a human sense without some tool to first translate and then send it to you.

It feels like a severe invasion of privacy. It is. But we've gotten super comfortable with cameras floating around all the time now.

It's only a matter of time before you just... It's like the phone, dude. The most influential got the phone first, mobile phones, smart phones, then started...

Then we all came on to it and now we can't live without it. Same thing will happen with the chip. Do you think you'll get addicted to the chip?

Absolutely. Like this conversation, imagine we're having it with a chip. You can have it in any world you want if it can write to you.

Have you seen Cyberpunk? There's this TV show called Cyberpunk. It's based on the game.

And it's like this thing. What is sold at that point in the world is skills. You can log on to this website.

It's all subscription. You can't buy it forever. $9 a month.

I now have access to Karate. I can do whatever. Whoever's selling it, he's put the exact sequence you need to download.

So, we don't know. That was crazy. If I put a scenario where AI enabled human versus AI enabled robot.

Who will win? Robot. Plus you have a self-preservation thing.

I will make sure I don't get hurt in the knees. I'll be able to run faster. No, but you have self-preservation.

Like a robot is okay tearing off its leg and hitting you with it for a weapon. You can't do that. Moron.

You won't have a leg after this.

[Speaker 4] (1:45:52 - 1:45:53)

What's wrong with you?

[Speaker 1] (1:45:53 - 1:46:40)

I have the chip in my brain which will like that game. I'll figure out something else. But I'll have to die 100 times.

Because our brain can talk to us. If you slit your thumb, let's say, it takes like quarter of a second till the pain kicks in. That's delay.

I don't think a robot has that delay. So, a robot can move much faster than you. It doesn't have pain.

At what point do you cease to be a human? Like in this scenario, if you're responding to a chip. I mean, as the most rudimentary electrical system, right?

It's sodium, potassium, calcium. It's the best system on the planet. You know, if Appu was a robot and you said, sing it.

He would sing it. I know.

[Speaker 5] (1:46:41 - 1:46:41)

I wish.

[Speaker 2] (1:46:44 - 1:46:57)

But we've watched all these science fiction movies. Which one has called the future correctly in your opinion? The next 10 years.

Do you think we will plug our brain into a machine and... There is a movie called Upload.

[Speaker 1] (1:46:57 - 1:46:59)

Sorry, Grid. Have you seen it?

[Speaker 5] (1:46:59 - 1:47:00)

No. Who's in it?

[Speaker 1] (1:47:06 - 1:47:10)

It's about this guy who gets paralyzed. Actually, I'm not going to spoil the movie for you. It's a must watch.

[Speaker 2] (1:47:11 - 1:47:22)

I remember Lucy. Lucy bought drugs. But at the end of the movie, she downloads her intelligence into a computer and she becomes like immortal.

[Speaker 1] (1:47:24 - 1:47:42)

There's a movie called Transcendence. In fact, in Transcendence, one of the first tasks the robot takes is how do I build myself arms? And it tricks a human into using...

You know what? ChatGPT can just hit the mechanical turk API. I don't know if it has an API in point, but whatever, it can create a listing.

[Speaker 2] (1:47:43 - 1:47:52)

So what do you say, up until it gets execution and it's only a chat GPT, then things are fine? No, it can execute already.

[Speaker 1] (1:47:52 - 1:47:55)

It can already hit any API. And now there are APIs to connect to humans.

[Speaker 2] (1:47:56 - 1:47:59)

But didn't we discuss earlier that ChatGPT can't execute, but AutoGPT can.

[Speaker 1] (1:47:59 - 1:48:45)

But someone uses an API. Okay. Even if, say, everyone had to ban AI.

You say stop. It can't. It's a Pandora's box thing.

That advancement has happened. Enough people have it in their computers. It's too late to stop.

And even if you stop open AI, there are open source models. They're not as good, not even close, but they're there. What is ChaosGPT?

To me, it just sounds like something. I don't know. I haven't read too much into it, but someone made a website saying here's what AI is capable of.

Imagine if here are all the scenarios that can cause absolute chaos.

[Speaker 2] (1:48:45 - 1:49:10)

So all these guys who are online critiquing the progress of ChatGPT for more, for slightly altruistic reasons, not for themselves to benefit in any manner, but thinking the world will end in one manner or the other if we don't check it. There are a lot of people doing that, right? Like take a six-month break, don't do it, this, that.

[Speaker 3] (1:49:11 - 1:49:12)

Elon, Yuval, everyone.

[Speaker 2] (1:49:13 - 1:49:18)

Do you think that community can weigh over? No.

[Speaker 1] (1:49:20 - 1:49:39)

I think it's the fear that somebody else will do it and rule them, which is scary. It's on Pirate Bay. There are open source models like Lama now on Pirate Bay.

But I think you can throttle… You can't. You can throttle things by shutting down GPU manufacturing.

[Speaker 2] (1:49:40 - 1:49:45)

Or can you throttle by making the internet less open-sourced in a way? No.

[Speaker 3] (1:49:47 - 1:49:52)

The other way of doing what you're saying is regulate. GPUs.

[Speaker 2] (1:49:53 - 1:49:55)

What do you mean by GPUs?

[Speaker 1] (1:49:55 - 1:50:09)

Graphic cards. Don't let Nvidia sell graphic cards. You throttle the compute and make five things you want and make math.

Destroy that whole hardware. Yeah, don't allow them to sell.

[Speaker 3] (1:50:09 - 1:50:33)

Think about it this way. Even governments can't regulate this because if you think about the birth of social media, which was in 2007, or 6 or even early 2000, really potent social media started in 2003, 4, 5, around that time. But 20 years since then, we don't even…

[Speaker 2] (1:50:34 - 1:51:06)

When they clone that sheep. That's a good point. So when cloning…

Now people can clone, right? In some version, they can clone. We all assumed that if the US bans it, China will begin to clone and they will have superior human beings and cloned beings and all of that.

But the crazy thing is China also understood the danger of this and the only person in the world who is in jail for having cloned somebody is a Chinese guy. And guess who put him there? The Chinese government.

[Speaker 1] (1:51:07 - 1:51:29)

Hey, but there's a very… It's two different things because with cloning, you need specialized equipment. To work on a model, you just need a computer.

And I don't think any computer is getting banned. Everyone already has one. You can work off

this.

Yeah, you don't even need a computer. How do you even process GPUs? Like we use…

Tell Nvidia not to sell them. Say if you want to sell it, get a license. If somebody wants to buy a GPU…

But are they used in just regular computers as well like a gaming PC?

[Speaker 2] (1:51:29 - 1:51:35)

But then maybe a government will use it versus somebody because somebody will figure out how to make a GPU like Nvidia at some point.

[Speaker 1] (1:51:35 - 1:51:54)

Very hard. Nvidia is the only company that's properly figured it out. AMD is still a…

But how will they know what GPU is being used for? Doesn't matter. You'll have to fill a license form like pharma.

You can't buy certain chemicals. Don't even have to use GPUs. You need a prescription sort of thing.

It's such a big thing, Nvidia's GPUs. It's not such a big thing. It's one way of rate limiting.

[Speaker 2] (1:51:54 - 1:51:57)

It's not blocking the GPUs.

[Speaker 3] (1:51:57 - 1:51:59)

Isn't that technology? Yeah. It is superb.

[Speaker 2] (1:51:59 - 1:52:02)

It's a company which gets like 40 multiples or something like that, right?

[Speaker 3] (1:52:02 - 1:52:05)

And Jerry is an amazing CEO. I don't know the finance part of things.

[Speaker 2] (1:52:05 - 1:52:09)

I would assume if they made this life-changing thing, they would be a lot more valuable, no?

[Speaker 3] (1:52:10 - 1:52:14)

Have you seen the stock over the last year? It's the only stock.

[Speaker 2] (1:52:14 - 1:52:19)

It's gone up a bit, but it's not like thousand percent, you know what I mean? I mean, not everything…

[Speaker 1] (1:52:19 - 1:52:24)

It may over the next two, three years. If everyone gets into AI, if every kid's getting into AI, maybe.

[Speaker 2] (1:52:24 - 1:52:25)

So, yeah.

[Speaker 1] (1:52:25 - 1:52:26)

But tell me this, okay?

[Speaker 2] (1:52:27 - 1:52:41)

We had industrial revolution, right? Like, where did it happen? Birmingham?

Yeah. Whatever. We had the factory assembly line and all of that.

Didn't we think then on similar lines to today that…

[Speaker 1] (1:52:41 - 1:53:04)

But you're not competing with human cognition there. You were never… You're competing with human output.

Like, as in work or efficiency. But I think the larger point Nikhil is trying to make is that people were being displaced from their job because of efficiency even before. That's true.

But newer categories… It's not really efficiency it's going after. It's thought.

As in, there was always a human behind the machine to control it.

[Speaker 2] (1:53:04 - 1:53:09)

You always needed a human. Less humans. But there was one human versus a hundred humans.

Sure.

[Speaker 1] (1:53:09 - 1:54:07)

Yeah. But this time that ratio is going to get really, really bad. Because most jobs…

But in a way it's similar. You know, in some ways. But I'll tell you why it's different this time, right?

And I know that people who make memes are saying that, oh, these AI people say it's different this time. You're going after cognition. You're going after the one thing that humans have that no machine ever had till now.

Like a steam engine couldn't think for itself and make decisions saying, should I go left or right? And that one person was still somewhere scared of getting killed or put into jail or, you know, something. But the machine doesn't.

Judgment is what's being replaced, right? Yeah. Judgment, thought, communication.

It's like, yeah, you'll have something running the steam engine but it'll be a human being, right? Now saying, will there be a human being run AI? Maybe for a short period.

But I think eventually you'll have AIs that can make decisions by itself. I'll show you something called Baby AI after this. You could fire your PMs tomorrow.

Don't do that, but you could fire your PMs tomorrow because it can give you the full spec.

[Speaker 2] (1:54:07 - 1:54:08)

PMs are what?

[Speaker 1] (1:54:08 - 1:54:38)

Product managers. Right? Because it'll tell you step by step what to do next for the next 36, 40 days.

Yeah. I'll show you. And it's really good.

Like, you want to build Slack, you can tell it, I want to build Slack, give me the steps. What's the name? It's called Baby AGI.

I'll show you. How do you get it? It's all free.

Yeah, it's all free, open source. You just need a open AI key, the GPT key. And it gives you every single step that you need to take and you can remove some of those steps if you want to.

So what I'm saying is much lesser human effort.

[Speaker 2] (1:54:38 - 1:54:42)

Can we all make a hypothesis of the next 10 years? We go one by one.

[Speaker 5] (1:54:42 - 1:54:43)

Sure.

[Speaker 2] (1:54:44 - 1:54:51)

How does the world change in 10 years in relevance to the conversation we've had just now?

[Speaker 1] (1:54:52 - 1:55:07)

I think, you know, broadly, everybody who knows how to use AI will be on one side, which will, they'll be rocking the world, right? On the other side, I think there'll be people who will appreciate, keep consuming, keep...

[Speaker 2] (1:55:07 - 1:55:12)

Why do you think people who know AI will be rocking the world? Because they themselves will be replaced in a way by the AI.

[Speaker 3] (1:55:12 - 1:55:14)

There'll always be a small percentage, right?

[Speaker 1] (1:55:14 - 1:56:27)

If it's, you know, 0.001% of the world, that's that, you know, percentage that will know how to prompt engineer, you know, all of those things. And they will be very, very relevant for the future. They will define how this world works.

They will define the rules of the new world. If you look at Instagram today, everybody behaves the same. So the consumers of the technology will all start behaving the same.

They'll all look the same. Like, on Instagram, everybody is doing the same poses, the same clothes, the same places they go to, you know. So, similarly, you know, these guys will define the rules, these guys will follow.

This whole podcast was a prediction only. This is what we've been trying to do, right? Instability will go up.

Instability. Offline. Offline will be at a premium.

Hopefully, there'll be a whole new category of things that will emerge where people are needed. That's the one thing that we haven't discussed, which is what are some potential new categories that will emerge? Like, when software happened, when mobile happened, there are a bunch of

new categories of workplaces.

[Speaker 2] (1:56:27 - 1:56:31)

That's actually a good question. If I'm 25 years old and I want a new job.

[Speaker 1] (1:56:32 - 1:57:31)

What are some new categories that are potentially open, Varun? Have you thought about this? See, if you go to a hotel, you know, there are things that AI can't do today or for a very long time.

It will not, like, from the check-in at the hotel, let's say you're going to marry at 5 star, to the room, the ambience of the room. Very tough for you to replicate the entire package. Say, until AI do everything, buy the real estate, put all this together.

Too many complicated tasks and too many offline tasks. So, I would say a lot of offline, new offline experience roles would come. Like, you're talking about the business, right, where people can get together and, you know, you have this group of founders or you have a group of creators or whatever.

Somebody being concierge for that group and, like, just setting it up and, you know, there's a bunch of roles that will pop up. You take the offline experiences of the US, right, versus a country like India, the hospitality in a hotel. Yeah, wildly different.

That's an opportunity.

[Speaker 3] (1:57:32 - 1:57:39)

No, they operate like a bot. Like, it's also because of really expensive labor, right? They just...

[Speaker 1] (1:57:39 - 2:00:01)

Yeah, but mentally also, if you go to a restaurant there and you say, I don't want this on my burger, but, you know, just say, remove something, they won't remove it. It'll take many years because of Moravec's paradox where it can't use fingers as well. For a few years, okay, like, I don't see a hotel...

Do you think prompt engineering is going to be a thing? What is prompt engineering? Is it English?

Yeah, but the intelligence to prompt... Dude, people are lazy, man. Like, they don't want to do anything, right?

I think clarity of thought is very important. See, my friend was telling me this, okay, he runs a design school. He's telling me that, do you know, he showed me a painting.

He's like, do you know what this painting style is called? I was like, I have no idea. He's like, if you wanted to prompt this, what would you say?

I would say, I said, okay, it's a painting with these kind of brushstrokes or whatever. He's like, it's Rembrandt. I don't know if I'm pronouncing it right.

He's like, it's Rembrandt. It's called Rembrandt style. And I was like, interesting.

Portraits. Yeah. Faces.

He's like, it's... I was like, nice. He's like, that's the thing about prompting.

It's like, you have a spell book. You can cast any spell you want and many people like you and me, because we've studied enough, we know the parts. We can use the spell book without opening the book.

A lot of people will need to. They'll need to look at somebody else's spell and be like, I'm going to copy that and make some tweaks to it. So, knowing these words, knowing history, knowing what would happen if X happened, I think those will be valuable.

Just your life experience. I think prompting is more an expression of your life experience and knowing what would happen. Like, if Nikhil was making a new app tomorrow, you were making another version of Koo tomorrow, you already know what works, what doesn't.

So, you'll be able to prompt it right, right, because you already know the mistakes. I think that life experience is what is valuable, not the English itself, because I think eventually everyone knows enough English. Again, if you have the chip in the brain, I can download any experience.

Oh my God, chip in the brain. Brian Naugat's tweet, right, which says, the industrial revolution rewarded the intensity of one's labor, the information age, the clarity of one's thought, and the AI revolution, the purity of one's taste. Yeah.

It's taste. The word is taste. Can you, can you use AI as leverage to bring to life the most accurate version of what you would imagine?

[Speaker 5] (2:00:01 - 2:00:02)

this is called?

[Speaker 1] (2:00:02 - 2:00:07)

Yeah. Do you know what this is, which is taste?

[Speaker 2] (2:00:07 - 2:00:08)

So, a lot more curation in everything?

[Speaker 1] (2:00:09 - 2:01:21)

I need to know the words, you need to know what that thing is called, because you can't conjure something that you don't know. You can give it a very lossy description that, oh, I'm looking for a game with this, that, or I could just say, I want a Valorant copy, right? Like there's this, there's this Instagram artist called Prateek Arora.

You should, you should check him out. He does this really amazing series called Indo-futurism, which is he imagines Indian culture, mixing that with what would be in the future. It's like such peculiar images.

Like he has a family, you guys should use the image by the way. He has this one picture called Indian family in the future peering through the Chakra view. Okay.

And it's one Indian family looking at this round, you know, globular structure, you know, almost like, what is that Veritasium in Harry Potter, like it's going to tell you about the future, you know, which is it's purity of his taste. Like he's found such an interesting thing, which would not, just because someone can use mid-journey, they wouldn't be able to. But there's a worry with that, that I have with that.

Sure, you could have fantastic purity of taste, but once you've proven it, if I'm a larger company, I can just copy paste. Yeah.

[Speaker 2] (2:01:22 - 2:01:32)

That's my worry. Yeah. What's your version next 10 years?

Painter, Orwellian, dystopian, Tanmay's version of 2033?

[Speaker 1] (2:01:33 - 2:01:39)

Oh man. What if he just chose to log off all this and live in Goa?

[Speaker 2] (2:01:40 - 2:01:51)

I'm already starting, I'm planning a fruit business. I'm going to grow fruits and sell them. Fruits.

Yeah. Exotic fruits like Rambustine, Mangosteen, Rambutan.

[Speaker 1] (2:01:52 - 2:04:29)

Yeah. I think, like the positive sides of AI, which is what I like to talk about generally, with some of my friends, right? Like I have, I've worked with colleagues and where I'm already trying to tell

them, say, hey, leverage AI, use it, use it forever.

And then that gets me thinking that, hey, listen, the tools, the tools have gotten better and better over the last decade. But the kids, tools are in abundance, but the curiosity is what is actually missing, right? So I fear a general feeling of like whatever social media did to the next generation's brain, AI will probably accelerate it.

And that is kind of worrying, which is, I think envy will go through all time high for those who are able to leverage it outside success for a lot more of your peers, which means you'll probably get way more envious and that'll make you, you know, react in such a way. That is my biggest concern, which is this next generation is is gonna be even more dopamine loaded, even more this thing. So tools have existed, but most people don't treat YouTube as a school necessarily.

So it'll just get more heightened with AI. So that, it's almost going to be Darwinian, like those who, those who choose the site to use it, it's that the results are going to be compoundingly higher. You know, on your point of dopamine, the frequency of dopamine hits have been increasing, right?

Like our parents used to have a regular life and then dopamine, once in a while when they went on a holiday or experienced something, they would get a dopamine hit or whatever. Then we got technology, we got addicted to more. As you said, I think AI will give us more dopamine hits per hour or per minute or whatever it is.

And then if we don't get it, even for a minute, it's like a drug here. Yeah, we're going to be addicts to AI, or AI generated stuff that, you know, if you're not part of it, if you're, once you're part of it, you're addicted. If you're not part of it, then you're going to go through a depression.

My inclination to learn more about AI just came out of pure fear. Like it was fear, then the more I started dabbling with it, it became a little more fascination. It's still fear.

Some of it is still fear.

[Speaker 2] (2:04:29 - 2:04:30)

Fear of missing out.

[Speaker 3] (2:04:31 - 2:04:35)

No, no, fear of fear of obsolescence.

[Speaker 2] (2:04:36 - 2:04:37)

And missing out.

[Speaker 3] (2:04:37 - 2:04:38)

Irrelevance.

[Speaker 2] (2:04:38 - 2:04:40)

Irrelevance, missing out, they are the same thing.

[Speaker 3] (2:04:41 - 2:04:44)

But missing out is too short term.

[Speaker 1] (2:04:45 - 2:05:34)

I'm saying it can change quickly if you start dabbling in it. The fear can turn into fascination more and more. The more comfortable you get with the idea.

Just don't have a rock brain, have a sponge brain and see how to go with it. I think that's a very good point. Everybody should be like that.

Varun? You know, the biggest driver of strife across history is a disgruntled elite or a fallen elite. And I think we're about to have that.

We're going to have these bunch of software engineers, white collar workers, just be out of a job, have to go back to being Uber drivers or like, there's nothing wrong in being any of those things or a petrol pump guy. There's nothing wrong with being any of these things, but these people will not like it. I used to be this, now how can you make me do this?

So I feel like we are white collar workers.

[Speaker 2] (2:05:35 - 2:05:38)

Because they have cash and capital, you know. How much?

[Speaker 1] (2:05:39 - 2:05:52)

Like, I don't think too much of India. Like, I read something about America. I don't know if it's true for India, but most remember the savings rate.

[Speaker 2] (2:05:52 - 2:05:58)

It's much higher. India and China are typically like, 20% per month.

[Speaker 1] (2:05:59 - 2:06:51)

Got it. So let's say that expires and that's going to expire sometimes. That's going to expire at some point.

I think they're not going to be happy and either way nobody likes losing a job. Even if you have a runway, it's like, yo, I had status, why did you take it away from me? Have you ever seen somebody gain status and then lose it?

They get desperate. You can see this with YouTubers, okay? Super relevant for like 6 years, 7 years, 10 years.

And then when they start dipping, they start going and creating the most extreme content. They're just like, please, I want to be relevant. Anything, I'll say anything, do anything.

Yeah, I'll say anything, do anything, slit whoever's throat just to be relevant. Every industry. So why do you feel that the elite in India, the white collar elites in India, I think they'll also do that at some point, right?

So my strategy is very simple, make enough money to build a giant wall, get my favorite people inside the wall, okay? Maybe 30-40 houses or whatever.

[Speaker 2] (2:06:52 - 2:06:53)

Is that why you're asking me about Goa?

[Speaker 1] (2:06:54 - 2:07:06)

Yeah. So basically we're all logging out and going to Goa. Maybe Trump had the right idea.

I know Trump said sound it for different reasons. He has cracked it. He's found his own country.

[Speaker 3] (2:07:07 - 2:07:12)

He's in jail now. A man is like, please edit this out.

[Speaker 2] (2:07:16 - 2:07:20)

Because if you have a scenario like this, I think a lot will sustain inside these walls.

[Speaker 1] (2:07:20 - 2:07:31)

I mean, we have to figure things out. It involves a lot of the thinking steps that you've been going through. That's why that conversation is fascinating to me.

Which is, how do we figure out energy? How do we figure out water? How do we figure out, you know?

[Speaker 2] (2:07:31 - 2:07:39)

I think for food, the best answer is not using the mediums you're used to, but something like

hydroponics.

[Speaker 1] (2:07:40 - 2:07:42)

We'll have to figure out how to farm in a smaller space.

[Speaker 2] (2:07:42 - 2:07:55)

You can actually. So in my house in Mumbai, it easily produces enough for the family.

[Speaker 1] (2:07:55 - 2:08:44)

For the family. Yeah, that's a good thing, right? And it's just like, my worry is like, people will end up getting very violent, the law is going to be full, they will not be able to handle so many instances of violence.

People are just not making money, right? Like, you look at a state in India that's not making that much money, You have similar issues, but at a smaller scale and the law is overburdened, it gets even worse, right? So I'm just going to protect myself and protect the people I love.

I know that sounds very selfish. No, no, it's very happy. Happy sounding.

Few minutes. It's been the one guy who understands the world. Look at Sam Altman.

Sam Altman is like, he said this in an interview. Imagine he runs OpenAI and he said this. I've kept guns, gas masks, gold.

Potassium iodide. Gold, guys.

[Speaker 2] (2:08:46 - 2:08:49)

Potassium iodide. Something that resonates with me. And antibiotics.

[Speaker 1] (2:08:50 - 2:09:11)

Not jewellery. And antibiotics. He's like, I just have these six things.

I put it in a package and that's going to be my... Can you imagine the CEO of any bloody company at that scale talking like this, saying my tech is going to be so blah, blah, blah, that I need all these things, you know, as my... Like, I don't know any CEO would say that.

Where did he say this? He said this on like an interview. It's public.

It's guns, gas masks, and something else.

[Speaker 2] (2:09:12 - 2:09:28)

A lot of people are making this use case for gold. Even because of, not wholly because of, but partly the banks failing, interest rates going higher, fractional banking, all of that, gold is seemingly a good idea.

[Speaker 1] (2:09:28 - 2:09:29)

How's gold performing in Zimbabwe?

[Speaker 2] (2:09:30 - 2:09:33)

Gold performance is universal.

[Speaker 1] (2:09:33 - 2:09:40)

It's a universal commodity. I fucking bought so much gold during zero interest rate phenomenon and I lost so much money.

[Speaker 2] (2:09:41 - 2:09:59)

But gold prices always go down when interest rates go up. I know, but I didn't understand anything, no. But right now, they're about $2,000.

So, if you want physical, generally, in most countries across the world, there's a premium. In Zimbabwe, because of inflation, that premium is probably higher.

[Speaker 1] (2:09:59 - 2:10:06)

You told me something last time we met. I don't know if you remember, but you said something along the lines of, if you have extra liquidity, invest in sovereign, something you said.

[Speaker 2] (2:10:07 - 2:11:22)

Sovereign gold bonds. You get like a 2.5% kicker. So, basic problem is we don't have any gold, okay?

India. We consume a lot of gold. I think...

Consume in what way? Import. So, when, say, my mother buys jewellery, if she buys a gold chain, that gold did not come from here, right?

It goes to different pockets of the world. It typically comes from Switzerland, Dubai, wherever. We import it.

Each time we import it, it's a deficit item for the country because we are getting foreign currency to buy that gold. To negate this... So, I think the number was 500 billion, something like half a trillion was the amount of gold we had imported in the last couple of years.

And to negate this, the government said, we will start gold bonds to add outside of the price of gold, which you will get the fluctuations and the increments. We will give you an additional 2.5% if you invest in SGV. Got it.

To negate the impact on the fiscal deficit that we are having. But it's a pity that we don't have gold because traditionally we did. And I think given enough research and enough time space to this, yeah.

[Speaker 3] (2:11:22 - 2:11:26)

But right now... We are right next to the only gold mine we have. KGFA.

KGFA.

[Speaker 2] (2:11:26 - 2:11:44)

But right now, a lot of central banks across the world, from China to Russia to all of these guys are starting to hold gold, which is a very interesting trend. And generally, following the smart money is a good idea. And smart money is big money and central banks are the biggest of them all.

[Speaker 1] (2:11:45 - 2:12:01)

So, that's why that math is very important to me, right? How much does it cost to build a wall and have these many people inside and figure out whatever? All the water, this thing.

And that can be a good driver for me. Worst case, if it doesn't happen. Then I've still generated enough money to do that.

[Speaker 2] (2:12:01 - 2:12:03)

But do you think that's the nicest way to live?

[Speaker 1] (2:12:04 - 2:12:06)

No. No.

[Speaker 2] (2:12:06 - 2:12:13)

No, absolutely not. If that's your assumption of tomorrow, today can't be happy. You know what I mean?

Yeah. I don't think it's...

[Speaker 1] (2:12:13 - 2:12:15)

So, I can do a little bit of detachment also.

[Speaker 2] (2:12:15 - 2:12:28)

Tomorrow, worst case, if shit happens, it's fine. We get buffeted by it. Like what we were discussing earlier, indifference is cool.

Yeah. You're going to die, maybe 10 years, you won't live, you'll die in 5 years. It's called...

There's this thing called indifferent attachment, okay?

[Speaker 1] (2:12:28 - 2:12:38)

Or distant attachment. Where you can live in the moment, you can be happy, you can have a long-term goal. But be like, you know, if I don't accomplish those long-term goals or things go to shit, it's fine.

[Speaker 2] (2:12:38 - 2:12:47)

I have a tattoo to remind me of that, see? There's no point, there's no tomorrow. Today is today.

And that's all there is.

[Speaker 1] (2:12:48 - 2:12:48)

Yeah.

[Speaker 2] (2:12:49 - 2:13:04)

Yeah, so... But like genuinely, like as a 30-year-old, so intelligent, like significantly more intelligent than I, an individual who's like, you know, has a wife and is probably going to have a kid and all that. I don't think...

[Speaker 1] (2:13:04 - 2:13:07)

Living in a state of pessimism is good. Yeah. Yeah.

[Speaker 2] (2:13:08 - 2:13:09)

I know.

[Speaker 1] (2:13:09 - 2:13:11)

He's not this bad unless you prod him. Yeah.

[Speaker 3] (2:13:13 - 2:13:13)

Prompt him.

[Speaker 2] (2:13:14 - 2:13:21)

We just prompt him. I have some friends who are like even worse. So, I know this.

Yeah. I deal with this. Like he laughs at jokes and all.

[Speaker 1] (2:13:22 - 2:13:25)

At a party, he'll smile. If you ask him to wave, he'll wave.

[Speaker 2] (2:13:26 - 2:13:42)

But imagine, like, just think, like, take yourself away from here and put yourself in 1939, when World War was happening. This is obviously much better. And everybody was dying at 30, or go to the Black Plague, or go to 1919, or go 200 years ago.

[Speaker 5] (2:13:42 - 2:13:43)

No, that I agree with.

[Speaker 2] (2:13:43 - 2:13:51)

Yeah, I agree with that. So, life is bad. The question is, do you fixate upon the negative or do you look for the positive?

[Speaker 1] (2:13:51 - 2:14:59)

No, I'm not looking at it from a negative. No, that's what I'm saying. I think this might happen.

There's probably a 10 or 5% chance it might happen. Like, I also know that the odds are not 100%. It's a 5% chance that it'll happen.

It drives me to the extent that, oh, I need to make enough money and have these things that are important to me in one place. It's a great motivating, like, a large-scale motivating desire for me, right, to do this, to build my own small mini-universe, I would say. And if it doesn't happen, cool, but at least I've worked towards it.

But if it doesn't happen, it's okay, right? 95% of the time, things are going to be okay. So, it's like, that's why I said, it's not doomed to the extent where I will make significant life changes.

It's just I have a long-standing goal and I'm like, okay, I'll try to hit that goal. So, 5% odds of this happening? Yes, 5%.

But 5% is enough of an odds to be worried about because you could have tangential paths, which

are not as bad, but still, you know, have this thing. I think either way, having community living is not a bad thing. Yeah, that I broadly agree with.

I think making a house for 25 of your closest friends. The only question is, is there a wall or not? Just to say, there are no walls.

[Speaker 3] (2:15:04 - 2:17:48)

So, my prediction is, I think none of this is going to happen. I think governments are smart enough at some point of time when they see this is a weapon of mass destruction, right? Now, you could argue whether the Nuclear Proliferation Treaty was fair, whether we were treated fairly as a country within it, whether, you know, even the United Nations is a functioning agency.

Do we have agency of our own? They can argue this for another limit, but I think there are two sides to it. And I will just abstract this from the way I look at it.

I think economically, the world will progress. India will progress. It's bound to happen.

Every country in the world is weakening. You could argue we're growing at 3%, 4%, or 5%, or 7%, whoever's data you want to believe. But we will economically progress.

Now, the thing that worries me is the point that you raised earlier, which is, are we creating enough employment? And are we creating enough sustainable employment, right? And to the point that you're saying that, is it dense enough?

Is it, you know, look at our cities. Our cities are crumbling, right? There's no infrastructure.

Everyone's moving from rural to urban. And that's not the way to live, or that's not the way sheer economic growth is going to happen. So, I think my prediction for the future in 10 years is, one, yes, this is a weapon of mass destruction, is the way I'll frame it.

Weapons of mass destruction around the world. There will be one example of a Hiroshima and a Pearl Harbor. I'm sure they will happen, to your point.

But I think it will soon come under curtail, right? There will be some proliferation that will come around the world to align, to cause extinction or any of this not happening. Having said that, I think the goal for leaders or the goal for whoever is setting into the next 10 years is, while we may be economically progressive today, but to maintain that demographic dividend, I'm hoping that we can create those sustainable jobs.

My view is, I think when we were born, we knew what our lifestyle of our parents were. And I can safely say the lifestyle we lead is significantly, when we started our careers, was way better. And that's a function of being beneficiaries to everything that we've seen.

Technology, mobile, growth, all of that. I think 10 years out, we will see that. But if we continue

down this path, where we don't care about the environment, we don't care about, you know, the point that you were raising is the uniform…

[Speaker 2] (2:17:48 - 2:17:49)

Universal basic income.

[Speaker 3] (2:17:49 - 2:18:52)

Yeah, universal basic income, a thought, right? I think we're going to have to change our thinking. But I don't think it'll change in 10 years, Nikhil.

I think it's going to take a little bit more than that. I think we'll see economic progress. I think we will see a better future.

And I'm optimistic that we will take some decisions to stop weapons of mass destruction. But more importantly, create a sustainable set of jobs. And to the threat that AI has, it does put a large threat on those jobs.

When I look at it, look at 2000, 2008. Every financial meltdown. And you're the expert here, right?

Why hasn't the Indian economy ever got dropped? You could argue it's because, good or bad, we've got a federal bank that is bloody strict. Even to do one stupid credit card transaction, now you have several OTPs and, you know, different…

You don't have an SVB happening here. I don't know.

[Speaker 5] (2:18:52 - 2:18:57)

I'm looking at you. I'm looking at you.

[Speaker 3] (2:18:57 - 2:19:00)

Just smile and wave, Nikhil. Just smile and wave.

[Speaker 2] (2:19:02 - 2:19:05)

So? No, no, I like the central bank.

[Speaker 3] (2:19:05 - 2:19:11)

You like the central bank. They sent me here to get that out of you.

[Speaker 2] (2:19:13 - 2:19:14)

No, no, I think…

[Speaker 3] (2:19:14 - 2:19:14)

That you shouldn't edit.

[Speaker 2] (2:19:15 - 2:21:02)

I think there are many acts that… I think the US should not have repealed a bunch of acts like the Dodd-Frank, Glass-Segel. They should not have recombined investment banking with normal banking.

All of that. Indian banking is a lot more conservative and the quality of Indian banks is probably superior. But if you were to look at a scenario where, you know, when the interest rates went up in the US, they went from a quarter percent to four and a half percent.

That rate of change is about 1,800 percent. Right? Long-dated paper in India works the same way.

We call them G-Secs, right? Government securities. Hasn't hit us as much because we have gone from five percent or four and a half percent to like six and a half percent, which is like a 20-30 percent change.

If a rate of change event like that occurs and six percent becomes 12, 15, 20, here too for everybody holding G-Secs, there will be mark-to-mark losses. And everybody has G-Secs. Banks, brokers, insurance companies, everybody.

I think this is something the government should probably not consider as cash. Today in the system, it's considered as cash. So, say, for example, you're buying a stock, right?

You buy 100 rupees worth of interest since we spoke about it. So, to get full margin from that stock, you need to bring in 100 rupees of cash. Right?

To create a pledge and get full margin. Today, G-Sec is considered as cash. I think that's a systemic risk.

[Speaker 3] (2:21:03 - 2:21:04)

They're using it as a tradable instrument.

[Speaker 2] (2:21:05 - 2:21:30)

Yeah, they're treating it as cash. And I think it's a long-term systemic risk in the system which has to be changed. Anything, any financial instrument which has a mark-to-mark component carries duration risk.

If I'm buying 20-year debt under any category, but there is a mark-to-mark element, interest rates

go up and down, things can go wrong.

[Speaker 1] (2:21:30 - 2:21:31)

Yeah, I mean, it's SVB.

[Speaker 2] (2:21:32 - 2:21:36)

Exactly, SVB can happen. It's just the rate of change is not the same.

[Speaker 3] (2:21:38 - 2:21:39)

Well, thank you for that.

[Speaker 2] (2:21:41 - 2:21:45)

But RBI rocks. RBI rocks. SEBI does too.

[Speaker 3] (2:21:45 - 2:21:47)

SEBI rocks. Yeah. There you go.

[Speaker 2] (2:21:47 - 2:21:59)

I think relatively, they're a lot more able, dynamic, and they change, and they are reciprocative of what people think compared to the rest.

[Speaker 3] (2:21:59 - 2:22:01)

Yeah, I agree. I would endorse that as well.

[Speaker 2] (2:22:01 - 2:22:11)

What do you think is going to happen in 10 years? Me? You know what being a stock trader 19 years full-time has taught me?

Nobody can f***ing predict the future.

[Speaker 1] (2:22:12 - 2:22:19)

I think. But you can have odds. You can have odds.

I can have odds. What's your optimistic prediction?

[Speaker 2] (2:22:19 - 2:23:05)

My optimistic expectation is productivity goes up across the world, across the board. UBI comes into play. UBI brings some kind of equanimity across the world.

Like I was saying earlier, I think capitalism works relative to everything else we've seen, but the version of capitalism we have probably has to change to a version where the anomalies, right? The people who have disproportionate rewards of capitalism, they have to become a bit more benevolent. It can happen either through tax slabs, inheritance tax, property tax, estate tax, stuff like that.

[Speaker 1] (2:23:05 - 2:23:07)

Did you tell this to Bill Gates when you met him?

[Speaker 2] (2:23:07 - 2:23:56)

Things like estate taxes and inheritance tax, right? There's so much precedent. Like if you go to US, if you're dying, you're giving your money to your kid, you pay 30-35%.

You go to UK, it's the same. You go to South Korea, you pay much more. But similar things happen in India and the richest folk in our country, they don't give anything.

And I don't think, I think entrepreneurship has to be encouraged, right? It's not like you're trying to make money, you're trying to scale a company, you're trying to work, all of that is fine. But if somebody you're competing with has predicated inherited wealth from 10 generations, that competition is not even.

So I think across generations, these taxes should exist. It will come. Yeah, I think it will come.

[Speaker 3] (2:23:56 - 2:24:00)

Estate tax is a part of becoming a developed economy.

[Speaker 2] (2:24:00 - 2:25:39)

And I think property tax is the best thing that will happen in India. Like if you were to live in, say, a pricey state in America, say Connecticut or something, right? You would pay as much as 2% of the value of your property every year as tax.

What happens to most of the ill-gotten wealth, black money in India, it goes and sits in real estate. There's no way of taxing that. For some reason, we don't have tax on farmers.

There are people like Monsanto, big agri-companies who earn 300-500 crores a year. They don't pay tax because they take advantage of that farm loan, right? There's a lot that has to be repealed in our tax laws.

But sorry, digressing, coming back to the point, I think productivity will go up. Capitalism will evolve in the manner that I said. UBI will come in.

I think UBI will create an ecosystem where people who want to pick capitalism can pick capitalism. Otherwise… Or play base-level socialism.

Base-level socialism. And I think those two exist hand-in-hand. I think, like, I don't know if this is true, but I have two people I work out with in the gym.

They just went on a holiday to New Zealand and came back, right? Husband, wife, married couple. They were telling me in New Zealand, most people do not care who you are, how much money you have.

It's a very… Equitable society. Yeah, it's a different society, but it works, right?

Like, you can choose to do what you want, he can choose to do what he wants. I think we'll go to a world like that.

[Speaker 3] (2:25:40 - 2:25:46)

Why are you so… So, question is, are you… Is this the idealistic Nikhil?

[Speaker 1] (2:25:46 - 2:25:48)

I was trying to paint the optimistic world.

[Speaker 3] (2:25:48 - 2:25:52)

He's an optimistic picture. Okay, so…

[Speaker 1] (2:25:52 - 2:25:53)

I want to get the pessimistic picture.

[Speaker 3] (2:25:53 - 2:26:10)

I want to ask you the question, which is, is this idealistic you? Which is the optimistic you? Or do you really believe…

Because what you said is compassionate capitalism. We all know that's the answer. We all know how hard it is to get there.

Do you really believe we'll get there in 10 years?

[Speaker 2] (2:26:11 - 2:26:43)

I think… Forced manner, I think you're saying. I think organically, right?

Like, it's funny I say this because I'm generally a fairly pessimistic person. But… I think

organically, people do right by people over the long term.

I'm not talking about the rich versus the poor, the gifted versus the ungifted, whatever. But society, humanity has lasted till this point and it has thrived because organically, we kind of do right by each other.

[Speaker 3] (2:26:43 - 2:26:46)

Interesting. So, you're not in the Machiavellian school of thinking, clearly?

[Speaker 2] (2:26:47 - 2:27:42)

No. I mean, see, I think Machiavelli, right? Like, I love his…

I mean, he's not got many books, but Prince, for example, or whatever. I think his viewpoints are very myopic, right? Like, he knew how to operate the Medici family, Italy of the time.

He knew how to get power amongst the bourgeois of that era. And, you know, his skill sets were limited to that. You can extrapolate and say Chanakya of Indian history or whatever.

So, I spent a lot of time reading random books, right? Like, different eras. But the one common trait, I think, is when push comes to shove, humanity kind of comes together and…

And we saw this with COVID recently. Yeah, I think so. It was a pessimistic picture.

I thought I mixed both of them a little bit. He brought up Machiavelli. Extremely pessimistic.

[Speaker 3] (2:27:44 - 2:27:50)

No, because I found it too idealistic. Yeah. And that's what I wanted to know.

That is it.

[Speaker 1] (2:27:51 - 2:27:58)

So, it's fine. There's been enough pessimism on this podcast. Yeah, I know.

It's fine. What a nice, soothing way to end. Yeah.

[Speaker 3] (2:27:58 - 2:27:59)

Yeah. Beautiful.

[Speaker 1] (2:27:59 - 2:28:00)

Yes. Love.

[Speaker 2] (2:28:04 - 2:28:05)

Who does that?

[Speaker 1] (2:28:06 - 2:28:08)

This is the Korean sign for…

[Speaker 2] (2:28:08 - 2:28:11)

What Korean stuff are you watching now?

[Speaker 3] (2:28:11 - 2:28:16)

Dude, I don't know. K-pop. Yeah?

I heard you say K-pop for a second, but, you know.

[Speaker 1] (2:28:16 - 2:28:20)

How is it a heart, man? This is a heart, no? It looks like a heart.

Oh.

[Speaker 3] (2:28:21 - 2:28:22)

Does it? Oh.

[Speaker 2] (2:28:22 - 2:28:23)

Like this.

[Speaker 3] (2:28:23 - 2:28:26)

Like this. Massive imagination. Oh, yeah.

Yeah.

[Speaker 2] (2:28:27 - 2:28:40)

So, I think that's it. Thank you all for watching, whichever camera is looking at me. Hi, I'm Nikhil Kamath.

I'd love to know what you thought of the episode. Comment, like and subscribe and thank you for watching.